

Distortions and Emissions: Fixing Markets vs. Taxing Carbon*

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Abstract

Optimal climate policy such as carbon pricing works by efficiently reallocating production to reflect social costs. India's electricity generation industry is a large and growing source of global CO₂ emissions, but unlike developed country markets, its marginal social cost curve is flat rather than upward sloping. I show that the subsidized allocation of domestic coal to dirty public sector generators is a key source of this distortion - public generators receive coal at 40% of international prices while efficient private generators must pay 2.5 times more despite showing 10–15% higher efficiency, resulting in \$4 billion in annual misallocation. Because the dirtiest producers are also the cheapest, subsidies cancel the carbon price signal before it changes which plants produce at the margin, leaving the observed MSC far from its efficient benchmark—a gap no standard carbon tax is large enough to close. I build an empirical model of India's electricity generation industry that captures generator incentives, binding long-term contracts, and grid operator optimization to estimate equilibrium quantities across generators and counterfactual policies. Coal price deregulation achieves CO₂ reductions equivalent to a \$40/tonne carbon tax while generating welfare gains 2.4 times larger (\$8.9 billion annually) and reallocating 20% of production from the dirtiest to the cleanest generators, compared to only 2.5% under the tax. At every tax rate from \$20–100/tonne, removing subsidies first would allow carbon taxes to achieve 130–270% greater emissions reductions, representing \$3.8 billion in avoidable annual climate damages. A Pareto improving reform combining deregulation with consumer transfers yields net welfare gains of \$1.9 billion. Findings suggest that in highly distorted economies, efficiency-enhancing reforms can serve as environmental policy itself.

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1 Introduction

Reducing greenhouse gas emissions is one of the defining challenges of our time. Standard economic theory prescribes a clear solution: tax emissions at their marginal external cost (Pigou (1932); Nordhaus (1993)). This Pigouvian logic underpins major climate policy architectures worldwide, such as the European Union Emissions Trading System, international carbon offset markets, and border carbon adjustment mechanism that together govern incentives for an increasing share of global production. Such policies work by efficiently reallocating production to reflect social costs, ensuring that for any given level of output, production allocates to producers that impose the lowest costs on society. India is a critical testing ground for this logic. It is among the world’s fastest-growing sources of CO_2 emissions, with coal accounting for 77.8% of total electricity generation and 40% of India’s carbon output, equivalent to 3.5% of global CO_2 emissions (Ritchie et al. (2020)). Coal-fired generation is estimated to cause 112,000 annual premature deaths from ambient $PM_{2.5}$ pollution (Cropper et al. (2021)). Given India’s projected economic growth and energy demand, decarbonizing its electricity sector is central to any credible global climate strategy (International Energy Agency (2023)).

Yet inside India’s electricity markets, carbon externalities coexist with severe pre-existing supply-side distortions that systematically misallocate production: large coal subsidies that favor entrenched public sector incumbents, interest subsidies that can accrue to state-owned enterprises or preferential allocation of inputs and contracts to politically connected firms (Ryan (2020a)). A rich literature documents that such distortions generate enormous efficiency losses across developing country industries (Hsieh and Klenow (2009); Restuccia and Rogerson (2017); Bergquist et al. (2026)). These distortions create a fundamental complication for carbon pricing: when pre-existing wedges systematically protect dirty, inefficient producers from market signals, adding a carbon tax does not reallocate production as efficiently as economic theory suggests. Several papers have documented how various market distortions reduce the effectiveness of first-best environmental policies (Shapiro (2021); Preonas (2024); Domínguez-Iino (2024)). Yet India has moved to introduce a Carbon Credit Trading Scheme (CCTS) that would include the electricity sector (Bureau of Energy Efficiency (2023)). Understanding whether first-best carbon pricing can be effective in this environment, or whether alternative policy reforms offer superior tools for emissions reduction is therefore both an empirically urgent and a theoretically important

question.

The central distortion in India's electricity sector operates through coal allocation. Coal India Limited, which supplies 84% of domestic coal, allocates subsidized domestic coal preferentially to public sector generators through long-term Fuel Supply Agreements (FSAs), at prices approximately 40% of international benchmarks. Many private generators must instead source from imports or spot markets, facing fuel costs 2.5 times higher, despite achieving 10-15% better operational efficiency (heat rates). Since a plant operating at 15% higher heat rate produces 17% higher carbon emissions per unit of electricity, this subsidy arrangement inverts the aggregate marginal social cost curve for generators: the plants with the lowest private (subsidized) costs are precisely those with the highest marginal social costs. These institutional features are common across capital-intensive industries in developing countries, making the findings broadly applicable. The economic logic connecting subsidies to carbon tax failure is direct: a carbon tax works by raising the relative cost of dirty production, shifting production toward cleaner plants at the margin. But in India, the observed marginal social cost curve is largely flat: the dirtiest plants are also the cheapest, and because their subsidy advantage effectively offsets the marginal carbon tax they face, this leaves the generators' relative position along the aggregate marginal social cost curve unchanged, and consequently, limited impact of a tax on emissions. The key theoretical condition is a negative covariance between subsidy size and marginal social cost across producers: firms that emit the most receive the largest subsidies, so the carbon price signal cannot overcome the subsidy wedge before it changes which firms produce at the margin. I show that removing the subsidy directly corrects this inversion, achieving larger emissions reductions and welfare gains than any carbon tax imposed on the distorted system. This finding departs from the second-best literature ([Lipsey and Lancaster \(1956\)](#); [Bovenberg and Goulder \(1996\)](#)), which takes distortions as given and asks what the optimal tax rate is. I show instead that distortion removal circumvents that question entirely, delivering welfare improvements that dominate carbon pricing.

Documenting that subsidies flatten or invert the marginal social cost curve is necessary but not sufficient to answer the policy questions at stake. Descriptive evidence establishes the negative correlation between subsidies and marginal social costs, but it cannot tell us how much distortions reduce welfare, what emissions would be under a carbon tax applied to the distorted system, or how production would reallocate if

subsidies were removed or redesigned. These are inherently counterfactual questions. The challenge is that observed generation reflects the equilibrium of an institutionally complex market, with long-term contracts locking plants into state-specific supply obligations, inelastic state-level demand, and daily capacity constraints, so that changes in generator costs under alternative policies would transmit through these frictions in ways that simple reduced-form estimates cannot capture. I address this through an empirical model designed to deliver three things: (I) direct measurement of private costs and coal subsidies at the generator level, (II) an estimated relationship between generator costs and equilibrium production across the full set of generators, and (III) the ability to conduct counterfactuals (carbon taxes or alternative subsidy designs) and re-solve for new equilibria.

The empirical model has a two-stage structure that mirrors how equilibrium generator-level production is actually determined. In the first stage, generators with power purchase agreements (PPAs), covering 88% of contracted capacity, decide whether to generate based on whether their marginal cost falls below their negotiated tariff; plants above the tariff withhold capacity. Merchant plants, those without PPAs, are not subject to state-specific constraints and compete in national merit order to clear the aggregate residual demand pooled across all states. In the second stage, the grid operator solves a cost-minimization problem daily, at the state level: state-specific demand must be met, plants cannot exceed available capacity, and PPA-contracted plants can only supply their dedicated states. This last constraint is empirically important: it prevents efficient reallocation of surplus capacity across state borders even when adjacent states face shortages. All components of marginal cost, fuel price, transportation, and operations and maintenance are directly observed in the data, so the model requires no cost estimation; I validate these estimates against variable costs that plants self-report to grid operators under their PPAs. State-level demand is treated as inelastic in the short run, supported by my demand elasticity estimates that are statistically indistinguishable from zero. Validated at daily plant-level generation across more than 200 plants and 1,000 days, the model explains 82% of the variation in actual output.

This structure delivers all three empirical requirements. For (I), plant-level coal sourcing and pricing data from 2018-2023 directly measure private marginal costs and subsidies separately at the generator level—no model assumptions required. The subsidy is observed as the gap between what a plant pays for coal and the market

benchmark price, observed daily. For (II) and (III), the empirical model provides the counterfactual mechanism: it estimates the relationship between generator costs and production shares across the full set of generators, daily and at the state level, conditional on the institutional constraints governing dispatch. Once this cost-production relationship is established, any policy experiment reduces to modifying generator costs and re-solving the model. Under a carbon tax, each plant’s marginal cost rises by the product of its emissions rate and the tax; under coal price deregulation, subsidized fuel costs are replaced with market prices. Because both instruments operate through the same channel of generator marginal costs, the model delivers structurally consistent comparisons between Pigouvian pricing and subsidy redesign.

My counterfactual analysis establishes that the current coal subsidy design renders standard carbon pricing largely ineffective and that the counter-intuitive second-best instrument is subsidy redesign, not a higher carbon tax. With subsidies intact, a carbon tax at any level from \$0–40/tonne generates welfare gains that are negative in the short run excluding environmental benefits, and achieves only 2.5% production reallocation across plants. The subsidy absorbs the price signal: state-owned plants’ access to coal at roughly one-third the market price offsets much of the carbon tax’s cost increase, leaving the relative dispatch ordering nearly unchanged. The punch-line is that a supply-side reform (coal price deregulation) outperforms the Pigouvian instrument at its own objective. Deregulating coal pricing achieves CO_2 reductions equivalent to a \$40/tonne carbon tax while generating annual welfare gains of \$8.9 billion, 2.4 times larger. This occurs because removal directly attacks the source of the blunting: it disproportionately raises costs for the highest-emission plants, who currently benefit most from subsidies, reallocating almost 20% of production from the dirtiest state-owned plants to the cleanest private generators.

The second part traces the mechanism through which the current subsidy design suppresses the Pigouvian signal. At every tax rate from \$20–100/tonne, combining with coal subsidy removal would allow carbon taxes to achieve 130–270% greater emissions reductions than they achieve in the status quo. The gap is not a matter of tax scale but of targeting: the subsidy advantage shields the dirtiest plants from the carbon price: subsidized coal at nearly one-third the market price offsets the cost increase from the tax, preventing it from changing which plants produce at the margin. More problematically, carbon taxes may accelerate clean generator exit during demand transitions. Marginal private producers with low capacity factors cannot

justify continued operation when the carbon tax adds costs but the subsidy keeps dirty incumbents artificially cheap: the tax eliminates clean plants first, precisely the opposite of its intended effect. The current subsidy design thus inflicts double damage on emissions policy—blunting the Pigouvian signal in the short run and distorting the exit margin in the medium run. At a social cost of carbon of \$190/tonne, this inefficiency represents additional climate damages of \$3.8 billion annually.

Finally, I simulate a politically feasible reform that deregulates coal pricing while protecting consumers through direct transfers. The policy allows domestic coal prices to follow international benchmark indices, eliminating fuel cost differences between plants sourcing domestically versus through imports. Simultaneously, the government compensates electricity consumers for resulting price increases, addressing the primary political constraint that originally motivated coal underpricing. This reform increases Coal India Limited revenue by \$8.4 billion while requiring \$7.8 billion in consumer compensation, yielding a net welfare gain of \$1.9 billion including emissions reduction benefits, demonstrating that Pareto-improving solutions exist even when addressing politically sensitive subsidies.

These findings have broader implications for environmental policy design in developing countries, where severe environmental degradation coexists with weak regulatory capacity ([Greenstone and Jack \(2015\)](#)). Unlike structural challenges such as building enforcement capacity, many production distortions are policy-created and thus more tractable to reform. Eliminating such distortions through unified policy reforms can serve as environmental policy itself, offering a more feasible path than traditional climate instruments. If outright removal remains politically infeasible, the efficient alternative is to redesign how subsidies are allocated. My theoretical framework implies a precise condition: efficient subsidy design requires that the subsidy received by each producer be inversely proportional to emissions intensity: the largest subsidies directed toward the cleanest producers and the smallest toward the dirtiest. The formal derivation appears in [Section 2](#). This is precisely the opposite of India’s current arrangement. Importantly, the Pareto-improving counterfactual (deregulation with consumer transfers) already approximates this efficient redesign directionally: when clean plants face lower costs they produce more, and the consumer transfer is channelled toward clean electricity consumption, reproducing the direction of the efficient subsidy allocation. The Pareto-improving reform thus provides a conservative lower bound on the welfare gains achievable from optimal subsidy redesign.

In terms of related literature, my paper makes three main contributions. First, by providing evidence that removing misallocation in production can dominate carbon taxes, widely considered the gold standard climate policy, I build on a rich body of work studying misallocation in developing countries. This literature has documented massive efficiency losses ([Hsieh and Klenow \(2009\)](#); [Restuccia and Rogerson \(2017\)](#)), and has identified state enterprise favoritism as an important driver ([Dollar and Wei \(2007\)](#)). I extend this work by showing that removing these same distortions can deliver larger welfare gains than direct environmental policies: deregulating coal allocation achieves CO_2 reductions equivalent to a \$40/tonne carbon tax while generating 2.4 times larger welfare gains. Unlike in efficient economies where carbon taxes appropriately target dirty producers, I show that when misallocation systematically protects dirty, inefficient firms, efficiency reforms emerge as the superior climate policy instrument. This finding follows tradition in the environmental literature in developing countries that has emphasized the importance of institutional distortions for environmental quality ([Greenstone and Jack \(2015\)](#); [Berkouwer and Dean \(2022\)](#); [Baylis et al. \(2023\)](#)) and success of environmental policy ([Duflo et al. \(2013\)](#); [Greenstone and Hanna \(2014\)](#); [Oliva \(2015\)](#); [Duflo et al. \(2018\)](#); [Jayachandran \(2022\)](#)).

Second, I extend second-best environmental policy theory ([Lipsey and Lancaster \(1956\)](#); [Bovenberg and Goulder \(1996\)](#)) by identifying when welfare gains from removing pre-existing distortions can dominate environmental policies. Previous work examines how market power affects carbon tax pass-through in developed countries ([Xia and Zhang \(2023\)](#); [Preonas \(2024\)](#); [Domínguez-Iino \(2024\)](#)), but has not examined distortions more common in developing countries such as administrative allocations and state enterprise favoritism. I provide an empirical framework showing that when pre-existing distortions create negative correlations between productivity and emissions intensity, carbon taxes lose targeting precision and may reduce welfare, making efficiency reforms the superior policy instrument. Carbon taxes may accelerate clean generator exit while protecting subsidized dirty incumbents, precisely the opposite of their intended effect, making efficiency reforms the dominant policy approach.

Third, I contribute to a rich literature studying supply-side distortions in wholesale electricity markets by identifying a novel source of inefficiency and connecting these distortions to environmental policy design. Previous work on electricity market inefficiencies focuses on cost-of-service regulation and market structure ([Fabrizio et al. \(2007\)](#); [Davis and Wolfram \(2012\)](#); [Cicala \(2022\)](#)), while recent work on India exam-

ines these same regulatory distortions (Ryan (2021a); Jha et al. (2021)). I extend this literature by identifying upstream market distortions, specifically administrative coal allocation, as a major source of supply-side misallocation, documenting \$4 billion in annual efficiency losses from fuel market distortions alone. Unlike previous work that focuses on production costs, blackouts, and reliability impacts, I am the first to connect electricity market distortions to environmental policy effectiveness, showing that removing upstream input distortions can achieve large emission reductions, and deliver larger welfare gains than direct environmental regulation. This finding suggests that the focus of the electricity market literature on downstream regulatory reforms may miss significant efficiency gains available from upstream market reforms, with broader relevance as distortions often transmit along the supply chain with aggregate implications (Acemoglu et al. (2012); Barrot and Sauvagnat (2016)).

The rest of the paper proceeds as follows. Section 2 explains the theoretical framework. Section 3 provides institutional background on India’s power sector and details the evolution of coal allocation policies and describes the data. Section 4 provides empirical evidence on coal market distortions. Section 5 describes the model and empirical strategy. Section 6 presents my main results, analyzing both the environmental costs of coal market distortions, and quantifying gains under counterfactual environmental policies. Section 7 concludes with policy implications.

2 Theoretical Framework

This section develops a formal framework to analyze when supply-side distortions (eg. coal subsidies) blunt the efficiency of carbon pricing and a sufficient statistic to compare their efficiency against alternative instruments such as coal subsidy removal. I demonstrate that the answer lies in the joint distribution of firm-level social costs and subsidies: when the correlation between them is large and negative, the carbon price signal gets significantly dampened, causing very limited reallocation in production. Subsidy removal turns more efficient and achieve larger emissions reductions and aggregate welfare gains than carbon taxation itself.

2.1 Illustrative Framework

Figure 1 illustrates the core mechanism through a simple supply-curve representation. In a well-functioning market, the aggregate supply curve for electricity production has two useful properties: it is upward sloping in private marginal cost, and—when

efficient production relies on cleaner technology—the social marginal cost (SMC) curve is also upward sloping. Panel (a) depicts this benchmark. Each step represents one producer; the blue staircase shows the private marginal cost (MC) curve and the red dashed staircase shows the social marginal cost ($SMC = MC + me$), where me denotes marginal emissions damage. Both curves are upward sloping, so the marginal producer at any aggregate quantity Q is both the highest private-cost and highest social-cost unit in the dispatch stack. A Pigouvian tax that adds $\tau \cdot e_i$ to each plant’s cost shifts the realized supply curve from the blue toward the red curve, re-ordering supply and routing production away from high-social-cost units.

Panel (b) shows what happens when coal subsidies that are directed towards high cost producers as in my setting, distort the supply curve. Generators are now dispatched in order of their subsidized private cost MC_s , which differs from true private cost by the subsidy received. The blue staircase is upward sloping by construction, since we sort by MC_s . The green dash-dot staircase traces $MC_s + me$ —the sum of subsidized private cost and marginal emissions damage. This curve is non-monotone. Some of the cheapest-to-dispatch generators carry large subsidies and are among the dirtiest; their high marginal emissions damage more than offsets their low MC_s , placing them at the bottom of the private cost stack but near the top of any social cost ranking. The result: the social cost ordering bears no reliable relationship to the dispatch order.

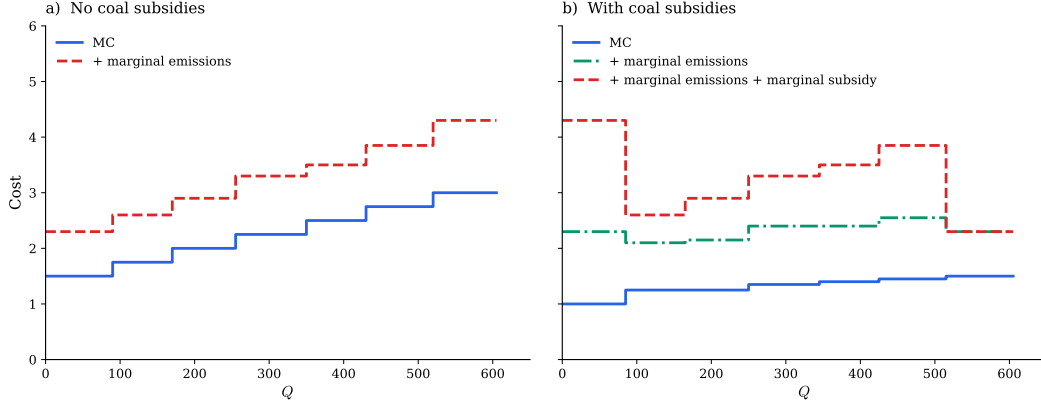
This unraveling of the social cost curve has a direct implication for carbon pricing. A Pigouvian tax at rate τ adds $\tau \cdot e_i$ to each generator’s cost, internalizing emissions and shifting the supply curve from the blue (MC_s) line toward the green ($MC_s + me$) line. The tax achieves efficiency gains only when this shift re-orders generators—routing production away from units with high social cost. But when the green curve is already flat or non-monotone, adding $\tau \cdot e_i$ proportionally to each generator does not change which producer is marginal and therefore, how much each generator produces. No meaningful emissions reduction are achieved—the channel through which carbon pricing generates its primary welfare gain.¹

Subsidy removal reverses this process. Moving from panel (b) to panel (a) eliminates the distortions d_i , shifting each generator’s cost back toward its true marginal cost c_i^* .

¹This logic is exact under inelastic demand, which we assume in the short run. With elastic demand, a carbon tax also reduces aggregate output, generating some welfare effect at the extensive margin. However, the reallocation channel—the mechanism through which carbon pricing improves productive efficiency—remains ineffective when the green curve is flat.

Because subsidies are largest for the generators with the highest social cost—removal disproportionately raises costs for the dirtiest units. The supply curve then realigns with the social cost ordering, reallocating production toward clean, efficient generators. The total welfare gain from subsidy removal exceeds what any feasible carbon tax can achieve in the distorted system. I formalize these results in the subsections below.

Figure 1. Marginal cost and social marginal cost curves with and without coal subsidies



Note: Each step in the supply curve represents one producer. The blue curve shows private marginal cost (MC in panel (a); MC_s in panel (b)), ordered by private cost. The red dashed curve shows social marginal cost ($SMC = MC + me$ in panel (a); $SMC_s = MC_s + me + \text{marginal subsidy}$ in panel (b)), where me denotes marginal emissions damage. The green dash-dot curve in panel (b) shows $MC_s + me$ —the sum of subsidized private cost and marginal emissions damage, excluding the subsidy component of social cost. The non-monotone shape of the green curve in panel (b) is the central result: when dispatch is ordered by subsidized private cost, social cost bears no systematic relationship to the dispatch ordering. Panel (a) resembles settings where marginal cost and marginal emissions are positively correlated, as in the United States where lower natural gas prices have made gas-fired generation both cheaper and cleaner than coal at the margin. Panel (b) captures India’s electricity sector, where coal subsidies systematically favor high-emission public generators, creating a negative correlation between private costs and emissions intensity.

2.2 Model Setup

Consider power plants i differing in emissions intensity e_i (kg CO₂/kWh). In an undistorted market, efficient plants have lower emissions, so $Cov(c_i^*, e_i) > 0$. Policy distortions create wedges d_i such that plant i faces operating cost $c_i = c_i^* + d_i$. When distortions are large and systematically favor dirty plants ($d_i < 0$ for high e_i), the correlation inverts: $Cov(c_i, e_i) < 0$. As a consequence, the distortions increase aggregate emissions. Figure 2 shows the distribution of generator-level costs and emissions

intensity underlying the two supply curve configurations in Figure 1: the left panels correspond to the undistorted benchmark of panel (a), where efficient generators have both lower private costs and lower emissions; the right panels correspond to the distorted setting of panel (b), where coal subsidies invert this ordering. Panel A of Figure 2 depicts the static cost-emissions distribution without a carbon tax. Each bubble represents a power plant, with horizontal position indicating marginal cost, vertical position showing emissions intensity, and bubble size reflecting capacity utilization. The region where dirty public generators operate is highlighted by red circles.

Under a carbon tax τ , plant i faces total cost $c_i + \tau \cdot e_i = c_i^* + d_i + \tau \cdot e_i$. Inelastic demand D is met by dispatching plants in merit order. To replicate the undistorted allocation, we need the cost ranking $c_i^* + d_i + \tau e_i$ to match the social cost ranking $c_i^* + \tau e_i$ —requiring an offsetting tax τ^* . The magnitude of this extra tax serves as a sufficient statistic for the efficiency loss arising from the distortions relative to the benchmark with no distortions. The magnitude also reflects the efficiency gains from removing the distortion itself - gains that accrue from both increases in efficiency of production as well as reduction in externalities.

2.3 Analysis

Static analysis of carbon taxation with supply-side distortions. Consider a social planner who seeks to minimize social costs (private costs plus environmental externality) while meeting inelastic demand D . Carbon tax revenues are collected and transferred out of the system (lump sum), so they do not enter the welfare calculation. The planner’s first-best problem is:

$$\min_{q_i} \sum_i (c_i^* + \tau_{social} \cdot e_i) \cdot q_i \quad \text{subject to} \quad \sum_i q_i = D, \quad 0 \leq q_i \leq \bar{q}_i \quad (1)$$

where q_i is generation from plant i , c_i^* is true marginal cost, e_i is emissions intensity, τ_{social} is the social cost of carbon, and $\bar{q}_i$ is capacity. This problem is solved by dispatching plants in ascending order of social marginal cost $(c_i^* + \tau_{social} \cdot e_i)$, which is equivalent to cost minimization.

In an undistorted economy, the market implements this allocation automatically: setting a carbon tax τ_{social} induces dispatch by $(c_i^* + \tau_{social} \cdot e_i)$, yielding the first-best.

With distortions, plants face costs $c_i = c_i^* + d_i$, where d_i represents the wedge from coal subsidies. Under carbon tax τ_{social} , the market dispatches by $(c_i^* + d_i) + \tau_{social} \cdot e_i$. The distortion d_i breaks the efficient production order and τ_{social} no longer represents the first-best policy.

Therefore, the planner chooses the optimal tax τ^* to approximate the first-best ranking. Setting $\tau = \tau_{social} + \tau'$, where τ' is a corrective component, the market dispatches by:

$$(c_i^* + d_i) + (\tau_{social} + \tau') \cdot e_i$$

For this to match the first-best ranking $(c_i^* + \tau_{social} \cdot e_i)$, we need:

$$d_i + \tau' \cdot e_i \approx 0$$

Since the planner cannot eliminate d_i , she uses the corrective tax $\tau' \cdot e_i$ as a second-best instrument to offset the distortion.

Optimal Corrective Tax.

The planner wants the market dispatch order under $(c_i^* + d_i) + (\tau_{social} + \tau') \cdot e_i$ to approximate the efficient order under $(c_i^* + \tau_{social} \cdot e_i)$. This requires minimizing the net distortion term $(d_i + \tau' e_i)$ across all plants. The planner minimizes total squared deviations:

$$\min_{\tau'} \sum_{i=1}^N (d_i + \tau' e_i)^2$$

First-order condition:

$$\sum_i (d_i + \tau' e_i) e_i = 0 \quad \Rightarrow \quad \tau^* = -\frac{\sum_i d_i e_i}{\sum_i e_i^2} = -\frac{\text{Cov}(d_i, e_i)}{\text{Var}(e_i)}$$

The optimal total carbon tax is $\tau = \tau_{social} + \tau^*$, where τ^* is the corrective component.

Economic Interpretation: What Does τ^* Measure?

τ^* is the corrective carbon tax component needed to restore efficient dispatch in the presence of distortions. The total optimal carbon tax in this second-best setting is $\tau_{social} + \tau^*$. The magnitude $|\tau^*|$ measures the social cost of the distortion - how much

extra carbon taxation (per ton CO₂) is required to overcome the misallocation caused by coal subsidies.

1. Measure of misallocation severity

$|\tau^*|$ depends on both the correlation between distortions and emissions, and the heterogeneity in emissions intensity. Large $|\tau^*|$ indicates substantial misallocation requiring a large corrective tax component to restore efficiency.

2. Threshold for carbon tax effectiveness

Consider carbon taxation at rate τ (on top of baseline distortions). Three cases emerge:

- If $\tau < \tau^*$: Tax insufficient to offset distortions; effectiveness severely limited
- If $\tau = \tau^*$: Tax restores efficient dispatch but achieves no additional emissions reduction
- If $\tau = \tau_{social} + \tau^*$: Tax achieves first-best

When $|\tau^*|$ is large, standard carbon taxes ($\tau = \tau_{social}$) fail to overcome the distortions that counteract the carbon pricing signal. Standard carbon taxes show limited effectiveness. Removing distortions ($d_i \rightarrow 0$, hence $\tau^* \rightarrow 0$) achieves the same reallocation as imposing a carbon tax τ^* , while avoiding the deadweight loss associated with taxation. Figure 3 below illustrates this threshold mechanism in the generator cost-emissions space. Each panel shows the distorted economy from the right panel of Figure 2 under a carbon tax τ that increases from left to right. As τ rises toward τ^* , the exit threshold progressively forces out the dirtiest generators, moving the realized allocation toward the efficient ordering shown in the left panels of Figure 2—and equivalently toward the supply curve configuration of panel (a) in Figure 1. The optimal subsidy design derived below achieves this same reordering directly, without requiring an elevated carbon tax to overcome the distortion.

Optimal Subsidy Design.

The corrective tax result has a direct implication for efficient subsidy design. Suppose political constraints prevent full subsidy removal ($d_i \rightarrow 0$ because for instance, they want to keep electricity prices low) but allow redesigning the allocation of subsidies

across producers. The condition $d_i + \tau^* \cdot e_i \approx 0$ implies that efficient design requires:

$$d_i \approx -\tau^* \cdot e_i$$

Subsidies should be *decreasing* in emissions intensity—the largest subsidies directed toward the cleanest producers and the smallest (or a net tax) toward the dirtiest. This is precisely the opposite of India’s current arrangement, under which the largest subsidies accrue to the most emissions-intensive public plants. The magnitude of τ^* —derived above as a function of the covariance between distortions and emissions—thus serves as a sufficient statistic for how far the current subsidy design departs from efficiency.

This result also clarifies the relationship between subsidy redesign and outright removal. If political constraints fix the total subsidy budget but permit reallocation across producers, efficient design achieves the same dispatch ordering as full removal: reallocating subsidies according to $d_i \approx -\tau^* \cdot e_i$ eliminates the misallocation without reducing aggregate subsidy expenditure. In the empirical analysis, I show that the Pareto-improving counterfactual (coal price deregulation with consumer transfers) directionally approximates this efficient redesign: when clean plants no longer face artificially inflated costs, they produce more, and the consumer transfer offsetting retail price increases is channelled toward clean electricity consumption. The Pareto-improving reform that I simulate in this paper is therefore a conservative lower bound on the welfare gains from optimal subsidy redesign.

Medium to long-run dynamics. While not the focus of this paper, I illustrate the medium to long-run implications of the theoretical mechanisms I discussed above. Electricity demand is elastic in this horizon and I allow generator exit. Let $\bar{C}$ represent the break-even revenue above which generators cannot profitably operate. $\bar{C}$ is also the electricity price or marginal cost of the marginal generator such that all generators on the supply curve to the right generate zero electricity. $\bar{C}$ is given by the demand, assumed to be elastic in the long-run. Generator i exits when total costs exceed this threshold: $c_i^* + d_i + \tau e_i > \bar{C}$

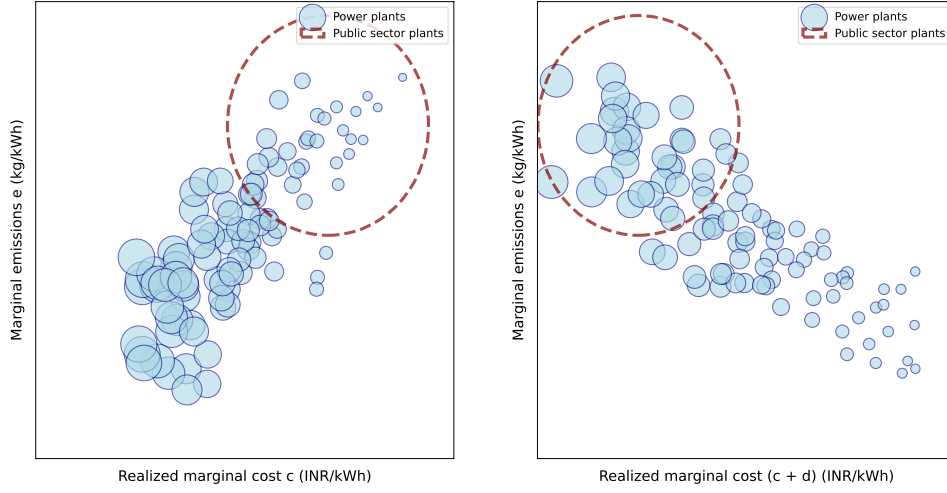
Panel B. of figure 2 illustrates the mechanism. The red line shows the exit threshold $c = \bar{C} - \tau \cdot e$, where generators above this line are forced to exit. Without distortions (left panel), the positive correlation between costs and emissions ensures that carbon taxation targets the correct generators—those that are both dirty and inefficient lie in

the upper-right region and exit first. This creates the intended environmental benefit through selective exit of the highest cost generators.

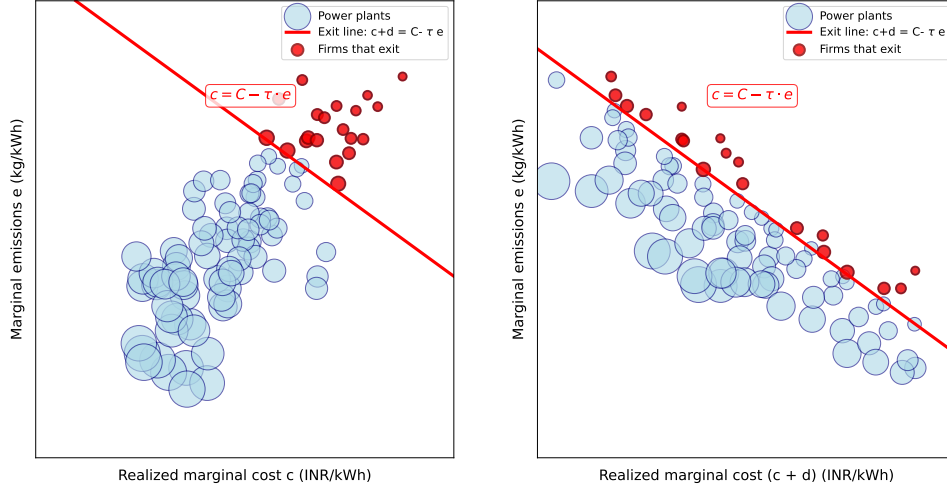
With distortions (right panel), since $Cov(d_i, e_i) < 0$, clean generators face high costs due to subsidies while dirty generators benefit from subsidies. The exit threshold cuts through both clean and dirty regions, causing adverse selection. Crucially, the distribution of bubble sizes in the right panel show that even among surviving generators, both clean and dirty plants maintain significantly high capacity factors, indicating that the carbon tax insufficiently reallocates production toward cleaner generators. This is because the existing distortions counteract the carbon pricing signal such that public generators facing subsidies are effectively shielded from the high cost of carbon taxation. At higher levels of carbon taxes, distortions are progressively negated. As established in the corrective tax analysis above, the allocation converges to the efficient first-best once τ reaches $\tau_{social} + \tau^*$ —at which point the realized cost ordering replicates the social cost ordering of panel (a) in Figure 1. Figure 3 traces this progression.

Figure 2. Carbon pricing with distortion

Panel A.

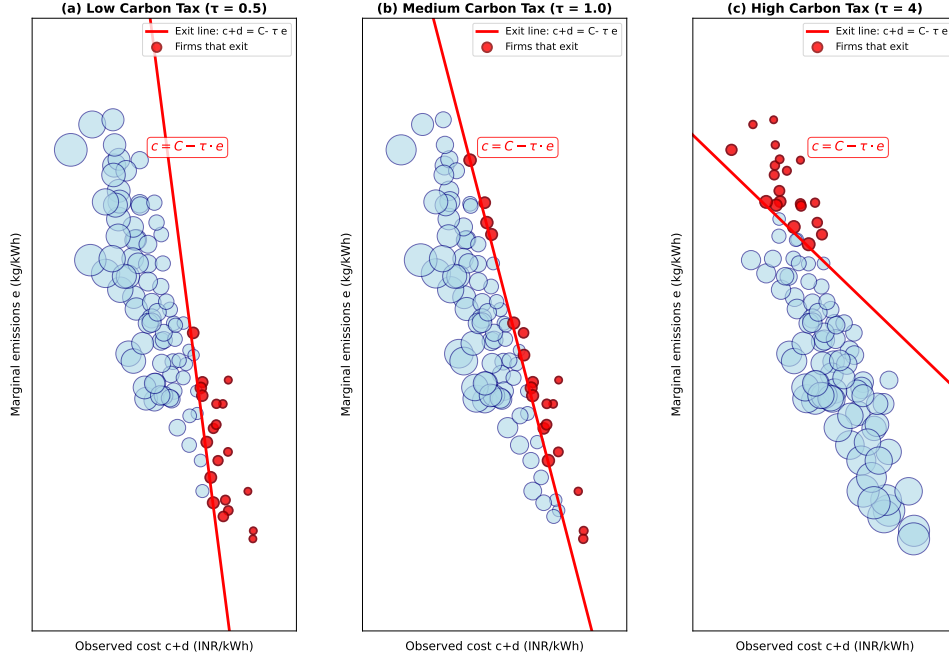


Panel B.



Note: c denotes generators' marginal operating cost in INR per kWh, and e denotes their emissions rate (kg CO_2 per kWh). Each bubble represents a power plant, with bubble size indicating the plant's capacity factor (fraction of available capacity utilized for generation). In panel A, the dotted red circles indicate the regions in which inefficient public generators operate. In panel B, the red line shows the exit threshold under a carbon tax τ , where generators with total cost $c + \tau e$ exceeding the break-even revenue C are forced to exit the market (equivalently, $c > C - \tau e$). The left figures in both panels A and B show an undistorted economy where efficient generators tend to have lower emissions (positive correlation between c and e). The right figures in both panels A and B show a distorted economy where input subsidies create a negative correlation between operating costs and emissions, such that cleaner generators face higher costs due to limited access to subsidized inputs. Red dots indicate generators that exit under the carbon pricing policy.

Figure 3. Targeting under varying carbon taxes



Note: c denotes generators' marginal operating cost in INR per kWh, and e denotes their emissions rate (kg CO_2 per kWh). Each bubble represents a power plant, with bubble size indicating the plant's capacity factor (fraction of available capacity utilized for generation). All panels show the distorted economy from the right panels of Figure 2, corresponding to the subsidized supply curve of panel (b) in Figure 1. The red line in each panel shows the exit threshold under carbon tax τ , where generators with total cost $c + \tau e$ exceeding break-even revenue $\bar{C}$ are forced to exit. τ increases from left to right. The rightmost panel, where $\tau \approx \tau_{social} + \tau^*$, restores the efficient dispatch ordering of panel (a) in Figure 1. Red dots indicate generators that exit under the carbon pricing policy.

3 Context and Data

3.1 Dominance of Coal in India's Electricity Generation

Coal remains the backbone of India's electricity generation, accounting for 77.8 percent of total monitored generation from 2018 to 2023. This share has stayed remarkably stable despite significant renewable capacity additions and global pressure to transition away from fossil fuels. The growth of non-hydro renewables from 5.9 to 8.9 percent has largely come at the expense of gas generation rather than coal, with gas falling to just 1.8 percent of generation despite comprising 6 percent of installed capacity (Figure 4). Coal's sustained dominance reflects not just path dependence but the reality that renewable additions have primarily accommodated growing electricity demand rather than displacing existing thermal capacity.

India’s coal dependence stands in sharp contrast to trends in developed economies. While the United States has seen both the share and absolute level of coal generation decline substantially since 2010 (Appendix figure A1), India follows a trajectory similar to China’s earlier industrialization, with coal generation continuing to grow in absolute terms. This pattern shows little sign of reversing. The [International Energy Agency \(2023\)](#) projects that India will drive the world’s largest increase in energy demand over the next three decades, fueled by a 74 percent rise in urban population, tripling per capita incomes, and rapid industrial growth. Even with ambitious solar deployment, the IEA still forecasts a 10 percent increase in coal demand by 2050. Given these demand pressures, existing infrastructure investments, and the intermittency of renewables, coal will likely anchor India’s power system for decades to come. The question then is not whether India will continue using coal, but how to use it more efficiently while reducing environmental costs.

Figure 4. Electricity generation by source in India

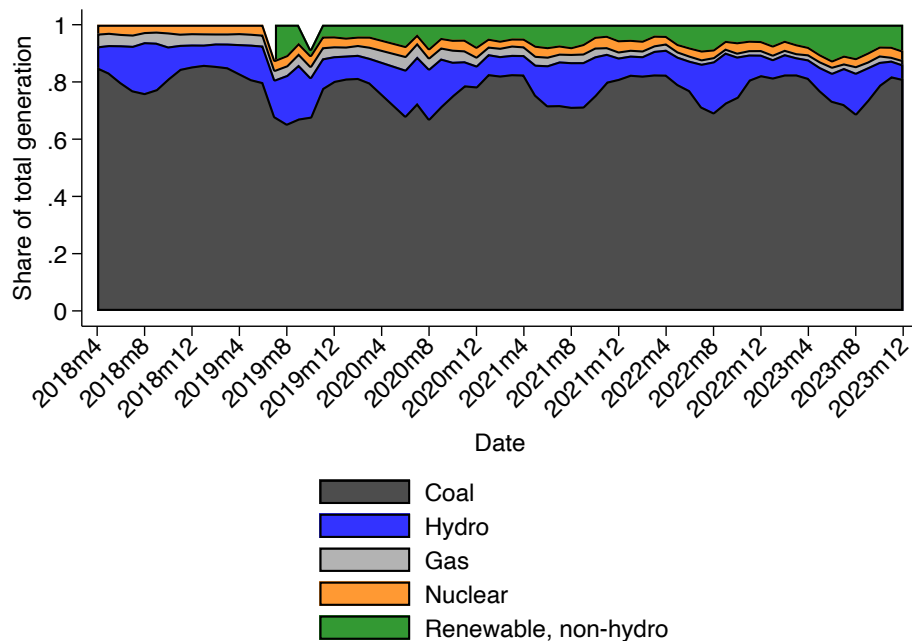


Figure plots monthly shares of total electricity generation in India by source from 2018 to 2023. The y-axis shows the fraction of total generation contributed by each source: coal (dark gray), hydro (blue), gas (light gray), nuclear (orange), and non-hydro renewables (green). Non-hydro renewable generation data are included from April 2019 onwards, when daily frequency reporting for these sources began. Source: [Central Electricity Authority \(2018-2023a\)](#)

3.2 Private sector participation in India's electricity generation industry

After independence in 1947, the Indian power sector was mostly publicly owned and run for more than 40 years. A sweeping economic liberalization in the early 1990s opened power generation to private firms, and the Indian Government solicited investment, including from foreign companies. For the power sector specifically, this period of liberalization was a failure. The deregulation of entry in generation reduced state control over generation, but retained the state's primary role in the transmission and distribution of electricity. As deregulation was not paired with any deeper structural reform, potential private entrants still faced monopsony buyers, the State Electricity Boards, in every state, and were therefore reluctant to invest ([Mathavan \(2008\)](#)).

Figure 5 shows the thermal generation capacity in the Indian electricity Industry from 1990 to 2019, and capacity by ownership (state and central governments together comprise public, and private). The privately-owned share of capacity is nearly zero (< 1 GW) until 2001, and is slow-growing thereafter.

Figure 5. Growth in installed thermal capacity by ownership

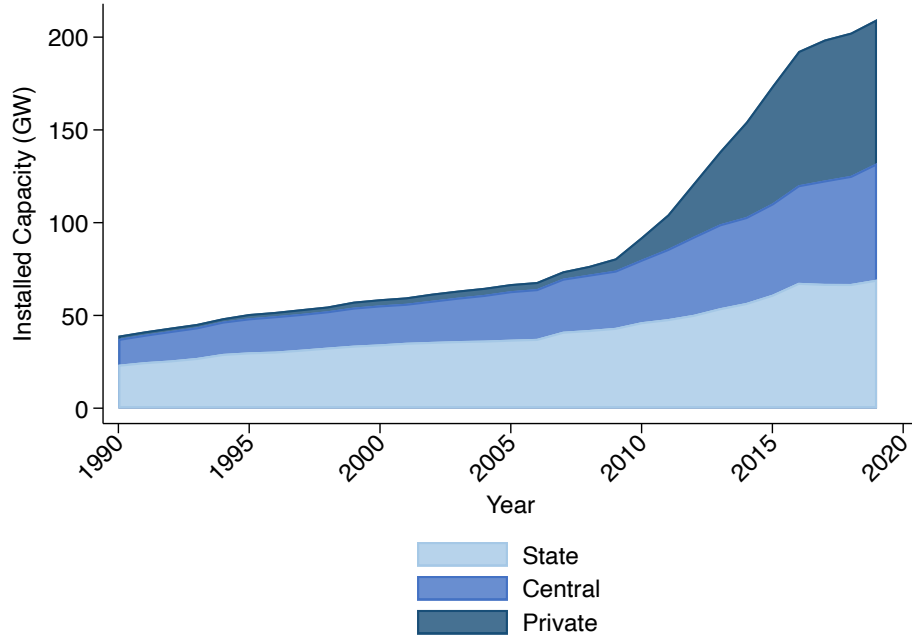


Figure plots the evolution of total installed thermal electricity generation capacity in India by ownership type from 1990 to 2019. The y-axis shows cumulative installed capacity in gigawatts (GW). Stacked areas represent cumulative capacity (in GW) for state utilities (light blue), central government (medium blue), and private sector (dark blue). Each area's height at any given year indicates the total capacity owned by that sector, while the overall height of the stacked areas shows total installed capacity across all ownership types. Source: Central Electricity Authority, India

The failure of this generation of private projects and power rationing built up pressure for deeper reforms. The Electricity Act of 2003 undertook structural reforms that recognized the natural monopoly nature of much of the power sector ([Kumar and Chatterjee \(2012\)](#)). Under these laws, the State Electricity Boards were separated into component parts, for generation, transmission, and distribution. Independent regulators were established to rule on power contracts and tariffs. Markets for power, though initially a small share of the sector, provided an outside option for private entrants to sell power ([Ryan \(2021b\)](#)). The National Tariff Policy (2006), issued under Section 63 of the Electricity Act (2003), mandated that state procurement of power must be done through a competitive bidding process ([Ministry of Power \(2006\)](#)). The auction results are formalized in a power purchase agreement (PPA), a contract between the supplier and procuring state utility or utilities that is written at the price set by the auction and reviewed and approved by the electricity regulator.

Several features of this procurement process made the industry attractive to private power sellers: the existence of Electricity Regulatory Commissions to approve contracts and arbitrate disputes; the principle of contracts being revealed at auction, which may lower contract prices and increase the transparency of price setting; the deference of the institutions now involved to the market process of setting prices; the large amount of power they are supplying, which may make them difficult to replace in the short term; and finally, the fact that the buy side consists of consortia of state bidders for large projects, which diversifies the political risk faced by sellers (Ryan (2020b)).

There was a large wave of private investment in power in response to the deeper structural reforms in the electricity sector and under the new bidding rules. The rules took effect in 2006 but the new projects built under these rules typically had a 4-5 year lag, meaning that they came online in 2010. Returning to Figure 5, we see a rapid increase in private generation capacity during this period, as shown by the blue curve, such that by 2019 private generation capacity was over 36 percent of the total capacity, greater than that owned by either the states or the Central Government. This dramatic transformation positioned private generators as major players in India's electricity sector, with many demonstrating superior operational efficiency compared to their public counterparts. Till the final date of this analysis, private sector continues to contribute 36 percent of the total capacity.

3.3 Coal Market Distortions in India's Power Sector

India's power sector is characterized by significant institutional distortions in coal markets that create systematic inefficiencies that distort both operational decisions of power plants and aggregate environmental outcomes. At the core of these distortions lies Coal India Limited's (CIL) near monopoly over coal production and distribution, coupled with regulatory structure that systematically favor public generators over private ones. These institutional arrangements misallocate resources across generators and undermine aggregate environmental performance by reducing the utilization of more efficient private plants while maintaining higher dispatch rates for older, less efficient public generators.

CIL produces 84 percent of India's domestic coal supply, with the remainder from Singareni Collieries Company Limited (SCCL) and captive mines. While CIL holds monopoly power, it doesn't exercise monopoly pricing. Instead, coal is primarily sold

through notified prices, which are administratively determined, grade-specific rates that implicitly carry subsidies. These prices are set to balance CIL’s profitability with the government’s objective of keeping coal prices low for end-users. Only 10-20 percent of total production is sold through e-auctions, where the premium prices reflect coal’s scarcity value [Ministry of Coal \(2014\)](#). Coal typically accounts for 60-70 percent of variable generation costs, making fuel access the primary determinant of plant competitiveness.

The real market for coal in India is essentially access to coal contracts or fuel supply agreements (FSAs) that specify supply guarantees at notified prices, but with explicit priority given to the power sector and further preferential treatment to public sector users. Those lacking secure access to domestic coal must rely on more expensive imports, creating a two-tier market where producer-level economic outcomes depend on administrative fuel access rather than efficiency.

3.3.1 Evolution of coal allocation policies

The institutional framework governing coal allocation evolved through successive policy revisions that systematically disadvantaged private entrants during the period of private sector expansion:

Pre-2007 Discretionary Regime: Coal allocation occurred through the Standing Linkage Committee (Long Term) [SLC(LT)], a body where public sector interests dominated—notably, the only consumer representative was the chairman of NTPC, India’s largest public power generator ([Sehgal and Tongia \(2020\)](#)). These allocations, particularly through Memorandum of Understanding (MoU) based arrangements, offered generous terms including high supply guarantees and favorable pricing. Most beneficiaries were public sector plants. The discretionary nature of these allocations faced legal challenges from private generators who were kept out, leading to Supreme Court intervention that mandated a more transparent, rule-based distribution system.

2007 New Coal Distribution Policy (NCDP): The NCDP introduced a structured framework with sector-wise priorities. While improving transparency, the policy focused primarily on ensuring coal availability to the power sector rather than enhancing distribution efficiency. Under NCDP, CIL established two main supply routes: long-term fixed price contracts through Letter of Assurance (LoA) leading to Fuel Supply Agreements (FSAs), and short-term price supply through domestic spot markets or e-auctions ([Ministry of Coal \(2007\)](#)). Private generators, being newer

entrants, predominantly fell under this regime with its stricter terms compared to the generous MoU arrangements that existing public sector plants had secured.

During the NCDP transition, CIL signed standardized FSAs with existing customers, promising 100 percent supply to the power sector; new projects, predominantly private, fell under the LoA-FSA mechanism. However, combined commitments under FSAs and LoAs far exceeded CIL's production capabilities.

2010-2013 Coal Supply Crisis: By 2010-11, as new generators came online, most of them private, CIL struggled to meet the growing demand for coal from these generators. Recognizing this supply constraint, CIL stopped issuing new LoAs. This particularly affected private generators who were still in various stages of project development. A 2012 Presidential directive initially mandated CIL to supply 100 percent coal for 78 GW of upcoming capacity under the 12th Plan (2012-17). However, facing supply constraints, a 2013 revision reduced this to 60 GW, with 11 GW receiving tapering linkages—temporary supply arrangements that decrease as plants' captive mines become operational.²³

Following the revised directive, the Ministry of Coal introduced differential trigger values (minimum supply guarantees)—90 percent for pre-NCDP consumers (mostly public) and 75 percent for new plants commissioned between 2009-2017 (predominantly private)⁴. These trigger values determine penalty thresholds for supply shortfalls, effectively creating different consumer classes based on contracting period. CIL linked contracted coal supply to long-term PPAs, prioritizing existing plants with firm commitments. While many private plants failed to convert their LoAs to FSAs due to strict milestone requirements, public sector plants faced fewer such hurdles. CIL continued to supply to meet demand with lower trigger values and those without firm FSAs, mostly private generators, were forced to rely more on imports.

Import-based plants and Policy-Induced Cost Shocks: Concurrent with domestic coal allocation challenges, private generators pursuing import-based strategies faced their own policy-induced setbacks that further undermined efficient generation. Several large coastal power plants, including Mundra Ultra Mega Thermal Power Project and Mundra Thermal Power Station, were specifically designed around im-

²The copy of the revised presidential directive is provided in Annexure VI of the CIL Annual Report, 2014: <https://d3u7ubx0okog7j.cloudfront.net/documents/2013-14'8w1zW4u.pdf>

³Define tapering/bridge linkage

⁴See Ministry of Coal's notification dtd. 26 July, 2013. https://coal.nic.in/sites/default/files/2020-01/260713_0.pdf

ported coal to circumvent India’s constrained domestic supply system. These plants represent some of India’s most efficient thermal capacity, employing supercritical technology and achieving superior environmental performance compared to older domestic coal-based units.

Import-based plants initially enjoyed significant advantages, with port locations eliminating India’s substantial inland transportation costs that can effectively double fuel costs for plants located 1000 km from mine sources. During competitive bidding in the mid-to-late 2000s, these plants offered tariffs competitive with domestic coal-based plants, making them attractive to state electricity boards. However, their competitive position deteriorated sharply due to exogenous policy changes. In 2014, the Indonesian government mandated that coal exports be priced at international benchmark rates rather than previously negotiated long-term contract prices, substantially increasing fuel costs for Indian import-based plants ([Taneja \(2013\)](#)).

This policy shift created a supply-demand mismatch where plants were locked into fixed-price Power Purchase Agreements based on original competitive bids but now faced substantially higher input costs. While many plants secured temporary compensatory tariffs—Mundra UMTTP and Mundra TPS received increases of 0.52 and 0.41 INR/kWh respectively—the Supreme Court struck down this practice in 2017. Till this day, these plants produce electricity with imported coal, paying international benchmark prices. The combined effect of domestic coal access barriers and import price volatility has left some of India’s most efficient thermal plants operating at reduced capacity factors, contributing to the systematic underutilization of cleaner generation capacity.

2017 SHAKTI Policy: The Scheme for Harnessing and Allocating Koyala Transparently in India introduced auction-based allocation for private generators while maintaining CIL’s monopoly structure ([Ministry of Coal \(2017\)](#)). SHAKTI closed the LoA-FSA route, requiring private sector plants without linkages but with PPAs to bid through discount offers on electricity prices. In stark contrast, the policy explicitly maintained preferential treatment for public sector plants, which continued receiving administrative allocations at notified prices without participating in any competitive process. This dual treatment within the same policy framework perpetuated competitive disadvantages facing private generators.

Access to Coal Deposits and Captive Mining: Beyond CIL’s direct coal supply, the evolution of captive mining policies created the most dramatic discrimination

between public and private generators. In 2007, amid shortfalls in domestic coal production and spikes in imported coal prices, industry appeals to the Prime Minister (who was also the Coal Minister) led to generous allocations of coal blocks on a first-come, first-serve basis at nominal prices, regardless of mining experience. These allocations later became controversial when the Comptroller and Auditor General identified an opportunity-based loss of Rs. 185,591.34 crore to the exchequer in what became known as "coalgate" ([Comptroller and Auditor General of India \(2012\)](#)).

The 2014 Supreme Court verdict cancelling 204 out of 214 allocated coal blocks created immediate financial distress for private power developers who had invested substantial capital based on securing low-cost captive coal. The cancellation eliminated low-cost coal supplies—increases from CIL's benchmark of 2,300 INR/tonne to 4,000 INR/tonne for market-sourced coal would raise variable costs by 1 INR/kWh, a 73 percent increase. Only 14 blocks were spared, including 12 allocated to NTPC and Sasan UMPP blocks?

The government's response through the Coal Mines (Special Provisions) Act, 2015, institutionalized the most explicit discrimination yet: mandating competitive auctions for private sector allocations while maintaining administrative "allotment" for public sector entities at nominal prices of INR 100/tonne for the power sector⁵. Private generators participated in competitive auctions with substantial premiums⁶—Monnet Power Company Ltd secured the Utkal C-block (with 123.86 million tonnes reserves) at INR 770 per tonne, excluding mining costs⁷. In contrast, public entities like NTPC received larger blocks at nominal prices, accessing blocks with geological reserves exceeding 7.3 billion tonnes. Most private blocks were sized under 10 MT, limiting economies of scale. By December 2018, only three private sector power plants were producing from their allotted captive mines([Central Electricity Authority \(2018-2023b\)](#)), highlighting the practical difficulties private generators faced in developing these competitively acquired assets and the limited success of the revised auction system in providing viable alternatives for private sector coal access.⁸.

⁵Economic Times (2015).<https://economictimes.indiatimes.com/industry/indl-goods/svs/metals-mining/psus-to-pay-rs-100-150-per-ton-on-coal-besides-upfront-payment/articleshow/45979075.cms>

⁶By August, 2015, 31 mines were auctioned off to the private sector, 9 of which were for power end-users

⁷Economic Times (2015).<https://economictimes.indiatimes.com/industry/indl-goods/svs/metals-mining/monnet-power-company-bags-utkal-c-coal-block-for-rs-9537-22-crore/articleshow/46505049.cms>

⁸These power plants include Sasan UMPP, Budge Budge TPS and Nigri TPP.

3.3.2 The Fundamental Distortion: Absence of Efficiently Functioning Coal Markets

These policy regimes - both in CIL supply and captive mining - have created systematic disadvantages for private generators through multiple mechanisms: timing of entry (post-2007 stricter regimes), explicit policy differentiation (public vs private treatment under SHAKTI and captive mining), and implicit advantages (composition of allocation committees, priority in shortages). Public sector plants receive administrative coal allocations with higher supply guarantees, better quality assurance, and priority during shortages. In contrast, private generators must participate in competitive auctions, face lower supply guarantees, and encounter more stringent performance requirements. This differential treatment forces private generators to rely on imported coal, significantly impacting their marginal costs and production levels.

3.4 Data

3.4.1 Electricity Generation and Outages

The Central Electricity Authority (CEA) publishes Daily Generation Reports, which include daily operational capacity and observed production for all utility-scale fossil, hydroelectric, nuclear and renewable power plants in India. I build an unbalanced panel combining the daily reports at the level of generator and day for the period of the study. The CEA's Daily Outage Reports provide the amount of capacity under outage for each plant on each day, except for non-hydro renewables. The reports also mention the reason for reported outage including fuel shortage.

3.4.2 Contract market data

I collect power plant capacity allocations to each state and the corresponding tariffs (both variable and fixed) for all power plants in the country from the Ministry of Power's MERIT portal. The portal reports the list of power plants contracted by the state utilities including the tariffs at which electricity was purchased from each. The dataset provides a complete description of the long term contract market including the tariffs (both fixed and variable costs) and contracted capacities.

The day-ahead market and short-term bilateral contracts comprise most of the short-term market. I use data on day-ahead market bids from the Indian Energy Exchange

(IEX). The IEX reports transmission region-wise demand and supply curves as well as market-clearing price and quantity at the exchange at 15-minute interval. The Indian electricity transmission network is composed of five transmission networks: North, Northeast, East, West, and South. Prices in the day-ahead market clear accounting for the transmission constraints between these five regions, resulting in region-specific prices. Only power plants with uncontracted capacity, and no fuel supply agreements (FSAs) with CIL participate in this market. I match thermal generators that participate in the market to the panel of generators above.

I also use short-term bilateral contract data collected from the Central Electricity Regulatory Commission (CERC). The contract data specifies the generating and buying entities, quantities traded as well as prices at the monthly level.

3.4.3 Wholesale electricity demand

I use daily data on the quantity of wholesale electricity purchased at the state level. The Power System Operation Corporation (POSOCO) operates the national grid and utilities report their electricity demand to the corporation daily. I utilise this data to construct a daily state-level panel of electricity demand.

3.4.4 Coal prices and quantities

The Central Electricity Authority’s (CEA) Daily Coal Reports provide plant-day level data on coal receipts, stocks, and consumption for generation. I complement this with CEA’s monthly coal reports that disaggregate receipts by source: Coal India Limited (CIL), Singareni Collieries Company Limited (SCCL), e-auctions, imports, and other sources. These reports also document monthly coal allocations specified in Fuel Supply Agreements (FSAs).

I construct source-specific coal prices using multiple data sources. For coal procured through FSAs (linkage-based coal), I use grade-specific notified prices from CIL and SCCL. Each plant is assigned a "minemouth" price based on the average grade (calorific value in kcal/kg) of coal it consumes. For e-auction coal, I utilize Ministry of Coal’s monthly Cabinet reports that document scheme-specific premiums over notified prices and quantities. I restrict to two primary schemes through which power plants procure coal: Spot E-Auction and Special Forward E-Auction. For months with missing data, I supplement with average scheme-specific prices and quantities from the Ministry of Coal’s Annual Reports.

For imported coal prices, I use monthly price series published by the Directorate General of Commercial Intelligence and Statistics (DGCIS). The DGCIS reports monthly quantity and value of steam coal imports (product code 27011920) by country of origin. I restrict to imports from Indonesia and South Africa as these two countries account for more than 90 percent of all imports. I validate them using a subsample of grade-specific coal import prices sourced from CoalMint.

4 Empirical evidence

This section provides both descriptive and reduced-form evidence that public sector generators receive higher coal allocation and receipts, pay lower coal prices compared to private generators and how that translates into a downward sloping marginal emissions curve. To provide empirical grounding to the structural model, I also provide evidence that private generators respond strongly to coal costs by reducing generation as the costs of coal procurement increase.

4.1 Efficient Private Generators Pay Higher Costs for Coal

Private generators show higher technical efficiency, measured in terms of heat rate (energy requirement for a power plant to produce one kWh of electricity). A lower heat rate means lower coal requirement per unit of output and hence, lower variable costs of electricity production. Appendix Figure [A2](#) reveals two key patterns in heat rates. First, there is a clear vintage effect - plants commissioned before 2009, predominantly public sector, operate at heat rates exceeding 2500 kcal/kWh. Second, and more importantly, private generators consistently achieve better heat rates than public plants of the same vintage. Among plants commissioned during 2009-18, private plants operate at heat rates around 2200 kcal/kWh while comparable public plants require about 2400 kcal/kWh. This efficiency advantage persists across all post-2009 vintages, reflecting private plants' adoption of superior technology.

The technical efficiency advantage translates into substantial economic benefits under competitive coal pricing. Private generators consistently achieve heat rates around 2200 kcal/kWh compared to 2400-2500 kcal/kWh for public plants of comparable vintage. This 300 kcal/kWh efficiency advantage translates to 0.24-0.37 INR/kWh lower fuel costs—representing 6.7-10.4% of typical electricity tariffs. More importantly for environmental policy, this efficiency difference directly translates to lower carbon emissions: public plants operating at 2600 kcal/kWh emit approximately 17%

higher CO_2 per kWh than private plants at 2250 kcal/kWh. Such efficiency differences would normally provide private plants with substantial competitive advantages, enabling them to underbid public competitors in both long-term and daily dispatch decisions.

Figure 6 quantifies this discrimination across vintage cohorts of power plants. Pre-2009 plants, almost entirely public sector, received generous coal allocations exceeding 350 tonnes/MW. Although allocations declined for all plants commissioned after 2009, public generators maintained a significant advantage over private plants. This allocation pattern aligns with the policy evolution described above - the 2012 Presidential Directive and 2014 cancellation of coal blocks cemented these differences. Notably, substantial private entry (shown by capacity bars) occurred precisely during these post-2009 vintages when coal access became more restrictive and discriminatory.

Figure 6. Coal Allocation and Receipts



Figure shows coal allocations and receipts by plant ownership and vintage. Left panel plots CIL/SCCL coal allocation (tonnes/MW) against plant commissioning year for public (red) and private (blue) generators. Right panel shows actual coal receipts in the same format. Lines show average allocation/receipt by ownership while bars show installed capacity (GW, right axis) in each vintage bin. Vertical dashed lines mark key policy changes: New Coal Distribution Policy (2007), Presidential Directive on fuel supply agreements (2012), and SHAKTI policy (2017), explained in detail in section xx.

Figure 7. Coal Receipts by Source

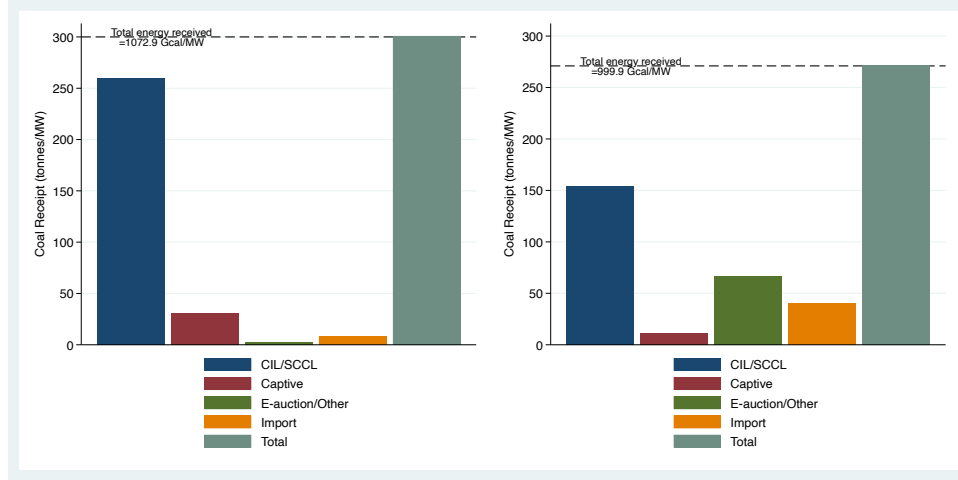


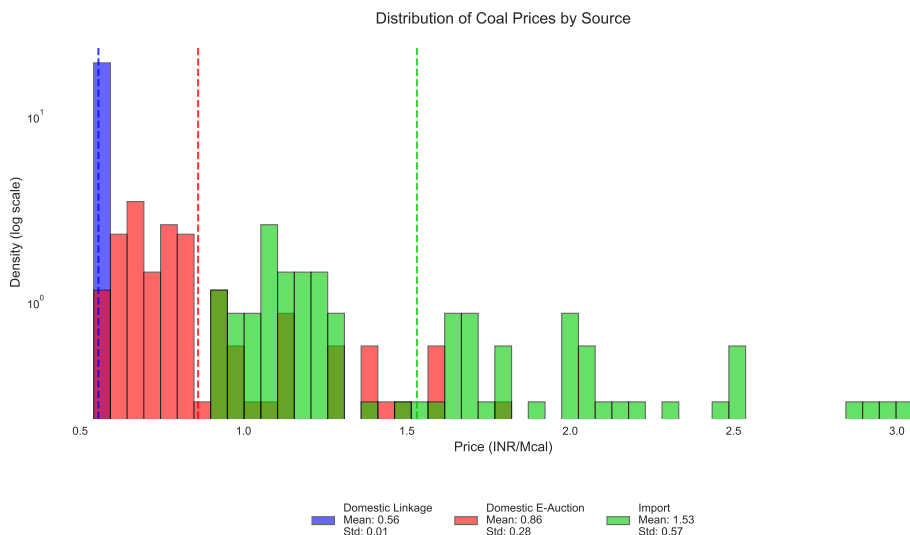
Figure shows the composition of coal receipts by source for public (left panel) and private (right panel) generators. Bars indicate average coal receipts (tonnes/MW) from different sources: CIL/SCCL linkage coal, captive mines, e-auction/other domestic sources, and imports. Total height represents aggregate coal receipts. Dashed lines show total energy received (Gcal/MW) after accounting for differences in coal quality across sources.

The right panel of Figure 6 demonstrates how these allocation differences translate into actual coal receipts. Public plants commissioned post-2009 receive around 250 tonnes/MW compared to about 170 tonnes/MW for private plants of the same vintage. Importantly, differences in receipts are far larger than differences in allocation, evidence that coal supply to public power plants is prioritised even when both public and private may have allocations with CIL. Figure 7 decomposes these receipts by source—public generators source most of their coal (over 250 tonnes/MW) directly from CIL/SCCL at subsidized prices, with minimal reliance on alternatives. In contrast, private generators receive significantly less CIL/SCCL coal (about 150 tonnes/MW) and must supplement with more expensive e-auction purchases, or imports. Even after accounting for the higher quality of imported coal, private plants receive significantly less energy per MW.

Figure 8 shows the cost implications of this supply discrimination. Expressing prices in energy-equivalent terms (INR/Mcal) to account for quality differences, private generators face substantially higher and more volatile input costs. Generators pay approximately 50% premiums over CIL notified prices for e-auction coal and 200% premiums for imported coal. During 2018-2023, an average private generator paid over 4500 INR/tonne for coal while typical public generators paid 2900 INR/tonne.

These price differentials create substantial cost disadvantages that overwhelm efficiency advantages. A thermal plant operating at 2200 kcal/kWh heat rate incurs fuel costs of 1.72 INR/kWh at CIL’s notified prices of 2900 INR/tonne. The same plant facing market prices of 4500 INR/tonne incurs costs of 2.68 INR/kWh—a 55% increase representing over 27% of typical electricity tariffs. Such cost differentials systematically reallocate production from efficient private plants introduction large productive inefficiency.

Figure 8. Distribution of Coal Prices by Source



The figure shows the distribution of coal prices by source during 2018-2023. The figure plots histograms of prices in INR per Mcal for three coal sources: CIL/SCCL linkage coal, CIL e-auction coal, and imported coal. Prices are computed at the plant-month level and expressed in nominal INR. The y-axis uses a logarithmic scale to accommodate the large differences in density across price bins. Vertical dashed lines indicate the mean price for each source. Refer to section 5 for a detailed description of the data.

4.2 Differences in Costs of Coal Introduce Large Productive Inefficiency

Under efficient allocation, plants with lower marginal costs should operate at higher capacity factors, as they lie lower on the supply curve. For plants receiving fixed tariffs through long-term contracts, profit maximization implies generating up to the point where marginal cost equals the contracted tariff. For plants participating in day-ahead markets, absent other frictions, lower marginal costs translate to lower bid prices and higher dispatch probability. Both mechanisms predict that more efficient plants—those with lower heat rates and, under undistorted coal pricing, lower marginal costs—should achieve higher capacity utilization.

Coal market distortions invert this relationship by creating artificial cost disadvantages for efficient private plants. The resulting generation pattern then systematically favors less efficient plants, increasing aggregate generation costs and emissions for any given level of output.

Figure 9 provides direct evidence of this misallocation by comparing plant utilization under current and counterfactual coal pricing. The left panel shows how current coal market distortions create a wedge between coal costs for public and private generators: higher-cost bins contain disproportionately more private capacity, and these plants - despite superior heat rates - operate at systematically lower capacity factors. The pattern directly contradicts efficient allocation where the most efficient plants should achieve the highest utilization.

The right panel quantifies the extent of misallocation by showing how plants would be reordered along marginal cost bins if all generators faced uniform pricing at CIL's notified prices⁹. Under this counterfactual, 40% of the cheapest capacity would be privately owned, compared to only 18% under current pricing distortions. The differences in capacity utilization between panels (solid red lines) reveal the magnitude of misallocation: the most efficient generators currently operate at capacity factors around 0.4 when efficient allocation would yield utilization rates of approximately 0.55. This analysis shows that moving from the current distorted allocation to efficient dispatch would increase output from the most efficient plants while reducing generation from less efficient generators, lowering aggregate generation costs and emissions for any given level of electricity production.

⁹The choice of CIL's notified price as the common price is illustrative. The key insight about misallocation holds for any common price, whether it's the international market price or the notified price. Import-based plants appear less competitive under current pricing partly because they don't benefit from the implicit subsidy in domestic coal, itself a distortion that affects dispatch order.

Figure 9. Misallocation in electricity generation

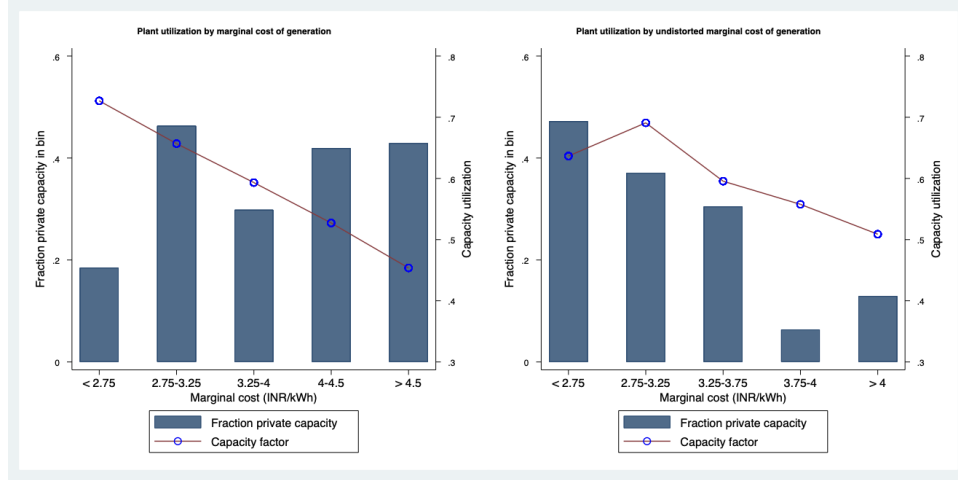


Figure shows the relationship between marginal cost of generation, ownership and plant utilization. Plants are grouped into cost bins (INR/kWh) on the x-axis. For each bin, bars show the fraction of total capacity in that bin that is privately owned (left y-axis) and the line shows average capacity utilization (right y-axis). Left panel uses observed marginal costs under current coal pricing. Right panel shows counterfactual marginal costs when all plants face CIL's grade-adjusted notified prices for coal.

The implication for the marginal social cost curve of this cost inversion is made direct in Figure 10, which plots the marginal emissions curve for Gujarat—a large demand center where public and private capacity compete within the same market. The vertical axis plots average emissions for typical marginal generators at every level of observed demand, measured as CO₂ emissions intensity (tonnes per MWh). Without current coal subsidies, this curve would slope upward: cheaper plants would also be cleaner, so a carbon tax would target progressively dirtier marginal units. Instead, the observed curve is steeply downward sloping: emissions intensity falls as we move up the supply curve to more expensive plants. The marginal generator—the one whose cost sets the carbon price signal—is substantially cleaner than the coal-subsidized inframarginal plants dispatched below it. This is the direct empirical counterpart to panel b) of Figure 1: coal subsidies de-couple the private marginal cost curve away from the marginal social costs, producing a marginal emissions curve that sloped downward rather upwards. The implication for carbon pricing follows immediately: a tax proportional to e_i cannot meaningfully reorder generation units when the marginal unit is already cleaner than the inframarginal ones it would displace.

Figure 10. Marginal Emissions Curve, Gujarat

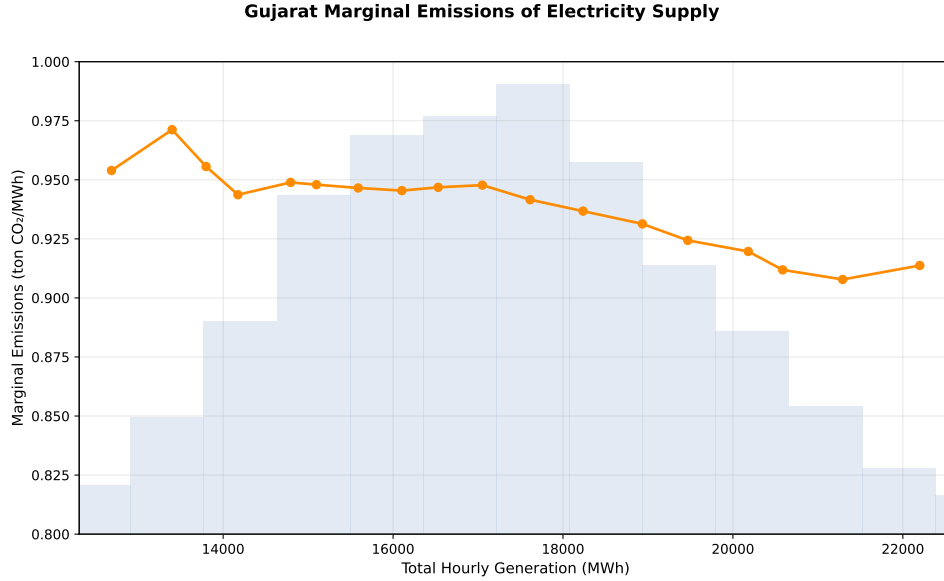


Figure plots marginal CO₂ emissions intensity (tonnes per MWh) against hourly electricity demand (GW) for the Gujarat electricity market, a large electricity demand center where public and private generators compete to meet demand. The background histogram shows the distribution of hourly demand over the sample period. Marginal emissions estimates are author's calculations from plant-level generation and coal consumption data.

4.3 Generator-level production responds to cost shocks

The observed misallocation from differences in capacity utilization between public and private plants could reflect several factors beyond coal market distortions. Private plants may face transmission constraints if they are located farther from load centers or have weaker grid connectivity, limiting their ability to supply power during peak demand periods. Technical factors such as different ramping capabilities, maintenance schedules, or plant availability could also contribute to utilization differences, particularly if private plant technology requires more frequent maintenance. Specifically to India, distribution companies (DISCOMs), which purchase most thermal power in India, face severe financial stress and payment delays that could influence procurement decisions—state-owned DISCOMs might preferentially purchase from public generators due to implicit government guarantees, political relationships, or simply because public plants are more willing to accept delayed payments given their own government backing. Additionally, differences in power contracting could matter: public plants typically have long-term contracts with assured off-take, while private plants may face greater exposure to short-term market volatility and demand uncertainty.

To establish that observed differences in capacity utilization reflect rational responses to cost distortions rather than other factors, I exploit variation in coal cost shocks both across ownership types (extensive margin) and within private plants over time (intensive margin). The coal price shock provides plausibly exogenous variation in input costs, particularly salient for private generators given their reliance on market-priced imports and e-auction purchases.

I estimate three specifications that progressively illuminate the cost-generation relationship. My baseline specification is:

$$PLF_{ist} = \beta_1(\ln CoalShock_t \times Private_i) + \beta_2 \ln CoalShock_t + \gamma X_{st} + \alpha_i + \delta_t + \epsilon_{ist}$$

where PLF_{ist} is the capacity factor of plant i in state s and month-year t , $CoalShock_t$ is the monthly price of imported coal in INR/tonne, and $Private_i$ indicates private ownership. The coefficient of interest β_1 captures the differential response of private plants to coal price shocks. I include plant fixed effects α_i to absorb time-invariant plant characteristics and month and year fixed effects δ_t to control for seasonal patterns and aggregate trends. The vector X_{st} includes state-level electricity demand to control for local market conditions. All specifications are weighted by plant capacity to account for heterogeneous economic significance.

To better reflect the economic costs faced by plants, my second specification (column 2) transforms the coal price shock $CoalShock_t$ into generation costs in INR/kWh. To do this, I convert coal price in INR/tonne to INR/kWh using plant-specific heat rates and the average calorific value of coal, which varies by month. This specification provides coefficients that directly measure how generation responds to changes in unit costs of production.

Finally, I examine the relationship between actual marginal costs and generation for private plants (column 3). The specification instruments the marginal cost of generation (in INR / kWh) with imported coal price to address the potential endogeneity in marginal costs. I restrict this specification to private plants given the very limited variation in public plant costs, which would otherwise pose identification challenges.

The fixed effects systematically control for various confounding factors. Plant fixed effects α_i absorb all time-invariant characteristics including technological parameters (design heat rate, vintage, capacity), locational advantages (proximity to coal mines, ports, or demand centers), and systematic differences in operational efficiency. The

inclusion of month fixed effects accounts for regular seasonal patterns in electricity demand, coal quality, and plant maintenance schedules, while year fixed effects control for secular trends in the power sector including changes in merit order dispatch, improvements in grid integration, and evolution of coal markets.

The two-way fixed effects specification addresses concerns that plant-specific operational decisions might be correlated with unobserved determinants of efficiency or that aggregate shocks might differentially affect plants based on ownership. For instance, if private plants are systematically newer or located closer to ports, the plant fixed effects ensure that our estimates of β_1 are not conflating ownership effects with these correlated characteristics. Similarly, the time fixed effects ensure that our estimates are not driven by concurrent policy changes or market developments that might affect public and private plants differently.

Table 1 presents strong evidence that private generators systematically adjust generation in response to fuel cost shocks, while public plants show minimal response. In the baseline specification (Column 1), a 10% increase in coal prices leads to a 0.95 percent decrease in capacity factor for private plants relative to public plants, representing a 1.6% decline relative to the mean capacity factor of 0.58. Public plants show an economically and statistically insignificant 0.11 percentage point increase, consistent with their insulation from market price fluctuations.

The pattern persists when measuring cost shocks in generation-relevant units (Column 2), with nearly identical estimates confirming that the relationship reflects genuine cost sensitivity rather than spurious correlation with coal price movements. The instrumented marginal cost specification for private plants (Column 3) reveals even larger effects: a 10% increase in marginal costs reduces capacity factor by 3.13 percentage points, likely reflecting the full extent of cost exposure including plant-specific efficiency and coal quality factors.

The systematic responsiveness of private plants to cost shocks suggests that observed differences in capacity utilization reflect rational economic responses to distorted input prices rather than technical constraints or other factors. The lack of response among public plants confirms their insulation from market signals, consistent with their preferential access to subsidized coal.

Table 1. Cost shocks and generation

	(1) PLF	(2) PLF	(3) PLF
log Coal price shock (INR/tonne)	0.011 (0.013)		
Private (=1) x log Coal price shock (INR/tonne)	-0.095*** (0.011)		
log Coal price shock (INR/kWh)		0.015 (0.013)	
Private (=1) x log Coal price shock (INR/kWh)		-0.094*** (0.011)	
log Marginal cost (INR/kWh)			-0.313*** (0.053)
Observations	11281.000	11281.000	4412.000
Dep var mean	0.580	0.580	0.573
Demand	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes
Month FEs	Yes	Yes	Yes
Station FEs	Yes	Yes	Yes

Table shows the relationship between coal price shocks and plant load factor (PLF). Columns (1) and (2) estimate the differential impact of coal price shocks on private plants using prices in INR/tonne and INR/kWh respectively. Column (3) estimates the direct relationship between instrumented marginal costs and PLF for private plants only. All specifications include plant, month, and year fixed effects, and control for state-level electricity demand. Regressions are weighted by plant capacity. Standard errors in parentheses are robust to heteroskedasticity.

4.4 Discussion - Implications for Environmental Policy Design

The descriptive and reduced-form evidence presented in this section establishes the key theoretical conditions that mute carbon tax signals: coal allocation distortions create a systematic negative correlation between the size of coal subsidies and marginal emissions across generators. Private plants achieve superior heat rates, which directly translates to lower marginal emissions, with efficiency differences of 300 kcal/kWh corresponding to 15-17% lower CO_2 emissions per unit of electricity generated. Yet, private plants face coal costs 55% higher. This means the cleanest, most efficient generators receive the smallest implicit subsidies (largest tax wedges), while the dirtiest, least efficient plants benefit from the largest subsidies—precisely the conditions under which carbon pricing systematically targets the wrong producers.

This negative correlation reduces carbon tax impacts through a clear mechanism. Under efficient allocation, carbon taxes increase the relative costs of dirty production, encouraging substitution toward cleaner alternatives and driving inefficient, high-polluting plants toward exit. However, when pre-existing distortions correlate negatively with marginal emissions, carbon taxes must overcome the large coal subsidies before meaningfully affecting the position of firms along the market supply curve. The coal cost advantages documented above can easily mute typical carbon price signals, preventing the reallocation of production toward cleaner generators that carbon pricing intends to achieve.

If distortions are sufficiently large, efficient private plants already operating at low capacity factors may exit entirely, while inefficient public plants with large coal subsidies face insufficient cost increases to alter their position on the aggregate supply curve. Rather than encouraging clean substitution, carbon taxation may accelerate clean capacity exit while preserving market share for dirty incumbents.

The mechanism suggests that removing the underlying coal allocation distortions may achieve significant reductions in emissions, even comparable with direct carbon taxation. Coal market reforms such as coal price deregulation that equalizes access to coal for all generators would naturally reallocate production toward more efficient, cleaner generators, without requiring difficult-to-enforce policies such as installation of mandatory abatement technology or carbon taxes. Over the medium to long run, with adoption of cleaner sources of energy and elastic electricity demand, the corrected incentives will cause exit of the dirtiest coal generators. Removing pre-existing distortions corrects incentives so that carbon pricing allocates production to reflect true

social costs and achieves emissions at levels society values. This makes efficiency-enhancing reforms a potential prerequisite for successful environmental policy rather than merely a complement.

The quantitative significance of this mechanism depends on the magnitude of current misallocation, the differences in marginal emissions across plants, and the size of distortions relative to the chosen carbon price. The following sections develop a structural model to simulate this mechanism and quantify whether removing coal allocation distortions can outperform carbon taxation at reducing emissions while delivering larger welfare gains.

5 Empirical Model and Estimation

5.1 Marginal Cost Construction

5.1.1 Coal-based generators

I estimate plant-specific marginal cost assuming that a plant's marginal cost does not vary with its level of output. To construct marginal costs of coal plants, I use the formula provided by the Central Electricity Regulatory Commission (CERC) Tariff Regulations 2019:

$$VC = \left(\frac{(SHR - SFC \times CVSF) \times LPPF}{CVPF} + SFC \times LPSFi \right) / (1 - AUX)$$

where SHR is the station heat rate (kcal/kWh), SFC is the normative specific fuel oil consumption (ml/kWh), CVSF and CVPF are the calorific values of secondary and primary fuels, LPPF and LPSFi are the landed costs of primary and secondary fuels, and AUX is the auxiliary energy consumption.

The primary component driving variable cost is coal consumption, determined by station heat rate and coal quality. I estimate heat rates from plant-level daily coal consumption and generation data, supplemented with annual reports of coal calorific values, reported by CEA. While CERC regulations specify normative values for other parameters like auxiliary consumption and secondary fuel use, I rely on observed heat rates since they vary significantly across plants and vintages. Plants source coal through multiple channels: fuel supply agreements (FSAs) with Coal India Limited (CIL) or Singareni Collieries Company Limited (SCCL), e-auctions, captive mines, and imports. The delivered primary fuel cost LPPF is therefore a weighted average

of source-specific coal prices, which I construct using:

- Grade-specific notified prices for FSA coal
- Auction premia over notified prices for e-auction coal (from Ministry of Coal reports)
- Import prices from DGCIS, adjusted for plant-specific reported coal quality
- Transportation costs based on rail distances from mines/ports to plants

I add government taxes, levies and surcharges to these costs to obtain the plant-specific “delivered price” of coal. For import-based plants, I assume zero inland transportation costs as they are located close to the coast and receive their supply via ships.

I validate our constructed variable costs against the Merit Order Despatch of Electricity for Rejuvenation of Income and Transparency (MERIT) database¹⁰ which reports costs based on power purchase agreements. For a consistent sample excluding lignite plants and those heavily reliant on imported coal, I find a correlation of 0.81 between my estimates and reported costs. More detailed methodology for construction of the marginal costs can be found in the Appendix.

5.1.2 Nuclear, gas and hydro based generators

Nuclear power plants in India have low variable costs, placing them near the beginning of the merit order. In the reported daily generation data, nuclear plants operate near capacity with an average plant load factor (PLF) of 0.96. Changes in the relative ordering of power plants along the supply curve in my counterfactuals do not affect nuclear utilization. Therefore, we place nuclear plants at the beginning of the supply curve to ensure they generate near capacity across all counterfactuals.

While hydro plants could theoretically serve as marginal generators during non-monsoon months (January through May and November through December), their actual dispatch reflects multiple constraints (Safiullah et al. (2017)). First, during monsoon months, generation is constrained by water availability. Second, hydro reservoirs serve multiple purposes including irrigation, drinking water supply, and flood control. Third, grid operators use non-merit order heuristics for hydro dispatch, as

¹⁰Government of India, Ministry of Power, Merit Order Despatch of Electricity for Rejuvenation of Income and Transparency (MERIT) (2023); <https://meritindia.in/>

evidenced by the strong correlation between hydro generation and net demand (demand net of renewables and coal generation) (Sengupta et al. (2022a)). To account for these operational constraints, we place hydro generators at the bottom of the supply curve alongside renewables and nuclear but constrain their generation to match observed daily levels. This approach captures both water availability constraints and periods when hydro plants may serve as marginal generators.

The treatment of gas-based power plants requires careful consideration due to significant data limitations. Plant-specific prices for domestic gas and imported LNG are not publicly available at monthly frequency. Plant-level data on gas transportation and marketing costs are also unavailable. Monthly-level fuel consumption data prevents reliable estimation of plant heat rates, particularly given the low capacity utilization of most plants. Moreover, a substantial share of gas capacity faces fuel shortages, as documented in CEA’s Monthly Gas Reports, complicating the estimation of available capacity in counterfactuals.

Given these constraints, I assume gas generation remains at observed daily levels in my primary analysis. This approach is justified by three factors. First, gas represents a small share of both generation (2.4%) and capacity (5.0%). Second, gas-based generation faces significantly higher costs compared to coal. Third, my main counterfactuals examining coal market distortions are unlikely to significantly affect gas utilization given the large cost differences.

The cost disadvantage of gas relative to coal is substantial. During 2018-2023, domestic gas prices averaged 335.9 INR/MMBtu (range: 130-764 INR/MMBtu), while long-term imported LNG contracts averaged 750 INR/MMBtu (9-10 USD/MMBtu). At a conservative 50-50 allocation between domestic and imported gas, the variable cost of electricity for an average gas plant is approximately 7.0 INR/kWh¹¹. This is more than twice the average variable cost of coal-based power plants in my sample (3.3 INR/kWh). In practice, the cost difference may be even larger as domestic gas supply is scarce and marginal requirements are typically met through spot LNG purchases. For instance, in April 2023, the lowest bid for gas-based peak summer supply was 13.7 INR/kWh, based on LNG pricing (Jain (2024)). At observed plant-level emissions factors, gas-based power plants become competitive with coal only at carbon taxes exceeding 100 USD/ton. Therefore, for carbon taxes between 0 and 100

¹¹These calculations follow standard assumptions from <https://www.fipi.org.in/assets/pdf/ep/Relative%20Costs%20of%20Power%20of%20Power%20Generation.pdf>

USD/ton, we expect minimal fuel switching from coal to gas.

I do not explicitly model generation from renewable generators as these have zero marginal cost and are therefore dispatched to available capacity. I set renewable generation to observed values in all our counterfactuals. With these assumptions, the model explained in the next section simulates generation using all dispatchable generators (coal, gas, nuclear, hydro and renewables).

5.2 Empirical Model of Electricity Generation

I build an empirical model of India’s daily electricity dispatch to deliver three empirical requirements: (I) direct measurement of private costs and coal subsidies at the generator level, (II) an estimated relationship between generator costs and equilibrium production across the full set of generators, and (III) the ability to conduct counterfactuals—carbon taxes or alternative subsidy designs—and re-solve for new equilibria. The model doesn’t rely heavily on assumptions around generator incentives or conduct, rather, it uses detailed micro-data and regulation to empirically represent actual generator incentives and the dispatch mechanism used by electricity grid operators, estimated and validated against daily plant-level data for over 200 plants and 1,000 days. No free parameters are estimated to fit outcomes: costs and capacities are directly observed.

5.2.1 Economic Environment

Let i index generating plants, s index states, and t index days. For each plant-day, I observe marginal cost $c_{i,t}$ (constructed as described in Section 5.1), maximum available capacity $\bar{q}_{i,t}$ (net of maintenance outages and equipment failures), and contracted capacity $\bar{q}_{i,s,t}^{PPA}$ for plants with long-term contracts with state s .

Market structure. Two mechanisms govern how electricity is bought and sold. The first is long-term Power Purchase Agreements (PPAs): state electricity distribution utilities (“discoms”) contract with individual plants, locking in dedicated capacity for 25-year terms. Each PPA specifies a variable charge—a fuel cost pass-through per kWh consumed—and a fixed charge covering capital costs. The variable charge is the operative constraint for short-run production decisions: a plant generates under its PPA only when its marginal cost falls below this contracted tariff. Since 2006, privately owned plants set variable charges through competitive bidding, with a share

indexed semiannually to a CEA-published coal price series. The second mechanism is a short-term spot market (power exchanges and bilateral contracts) that clears the remaining 12% of demand at competitive prices. I extract plant-state allocation fractions and variable tariffs (both fixed and indexed components) for each plant-state pair from the MERIT India database and Central Electricity Authority capacity allocation records.

The critical institutional feature of PPAs is *exclusivity*: contracted capacity is locked to its designated buyer state. A plant serving State A cannot redirect surplus capacity to State B, even if State B faces unmet demand and State A has excess supply. This constraint is not a modeling choice, it is the legal structure of the PPA arrangement, enforced through Long Term Open Access rights that guarantee transmission capacity only for the specified source-destination pair. When a contracted state’s demand falls short of contracted supply, the plant’s surplus capacity sits idle.

Demand. I treat state-level demand as perfectly inelastic in the short run. This assumption is supported by two observations. First, end-users face regulated retail tariffs adjusted only annually, creating a wedge between wholesale market prices and the retail prices that drive consumption decisions. The price signal that would generate demand responsiveness simply does not reach consumers in the short run. Second, I estimate demand elasticity directly and find estimates statistically indistinguishable from zero. Under inelastic demand, the model’s task is purely a supply-side allocation problem: assign a fixed quantity of generation across plants at minimum system cost, subject to institutional constraints.

Distortions enter through costs. Coal market distortions enter the model exclusively through generator-level marginal costs $c_{i,t}$. For plants with Fuel Supply Agreements (FSAs) with Coal India Limited, the fuel cost component of $c_{i,t}$ reflects subsidized domestic coal prices (approximately 40% of international benchmarks). For plants sourcing from spot auctions or imports, $c_{i,t}$ reflects market prices 2–2.5 times higher. Because the equilibrium generator level production depends directly on $c_{i,t}$, the subsidy structure shapes which plants generate at the margin. Counterfactual analysis then reduces to modifying $c_{i,t}$ and re-solving the same optimization: a carbon tax at rate τ enters as $c_{i,t} \rightarrow c_{i,t} + e_i \cdot \tau$, and coal price deregulation enters as $c_{i,t} \rightarrow c_{i,t}^*$, replacing subsidized fuel costs with market prices. Because both instruments operate through the same cost channel, the model delivers structurally

consistent comparisons across policy experiments.

5.2.2 Two-Stage Model Structure

The dispatch mechanism has a two-stage economic structure that mirrors the actual institutional process.

Stage 1: Generator availability decision. Each generator with a PPA decides whether to declare capacity available to the grid operator by comparing its marginal cost against its contracted variable tariff $\bar{p}_i^{PPA}$: the plant reports available if $c_{i,t} \leq \bar{p}_i^{PPA}$ and declares unavailability otherwise. When marginal costs persistently exceed the indexed tariff—as occurs for private plants that source coal from expensive imports or spot auctions—plants may seek PPA renegotiation. This withholding decision is the generator’s only short-run strategic lever; once capacity is declared available, the grid operator’s optimization determines dispatch quantities. Merchant plants (those without PPAs) always report capacity as available since they compete in the spot market at realized market-clearing prices rather than pre-committed tariffs.

Stage 2: System operator dispatch optimization. Given the set of available plants and capacity after Stage 1, the grid operator solves a cost-minimization linear program each day. The problem is:

$$\min_{\{q_{i,s,t} \geq 0\}} \sum_{i,s} c_{i,t} \cdot q_{i,s,t} \quad (2)$$

subject to:

$$\sum_i q_{i,s,t} = D_{s,t} \quad \forall s \quad (\text{meet state demand}) \quad (3)$$

$$\sum_s q_{i,s,t} \leq \bar{q}_{i,t} \quad \forall i \quad (\text{plant capacity}) \quad (4)$$

$$q_{i,s,t} \leq \bar{q}_{i,s,t}^{PPA} \quad \forall i, s \quad (\text{PPA allocation}) \quad (5)$$

Constraint (3) requires that each state’s demand is met exactly, reflecting inelastic demand and the grid operator’s obligation to maintain supply-demand balance at the state level. Constraint (4) limits each plant to its physically available capacity. Constraint (5) is the critical institutional constraint: PPA plant i can supply at most

$\bar{q}_{i,s,t}^{PPA}$ to state s —it cannot sell beyond its contracted allocation and, crucially, cannot serve states with which it has no contract ($\bar{q}_{i,s,t}^{PPA} = 0$ for non-contracted state pairs). Merchant plants face no such restriction ($\bar{q}_{i,s,t}^{PPA} = \bar{q}_{i,t}$ for all s), so the optimization freely allocates their capacity across any state with residual demand; these plants effectively compete in a national merit order to clear the aggregate residual demand pooled across states.

The solution to this linear program implements merit-order dispatch within the constraints of the contract structure: the cheapest available units are dispatched first, subject to state-level demand obligations and PPA exclusivity. The shadow price on constraint (3) is the market-clearing price: the marginal cost of the last (most expensive) unit dispatched to meet a given state’s demand on a given day.

Key friction: stranded PPA capacity. Constraint (5) generates an empirically important source of allocative inefficiency. When a contracted plant has available capacity beyond what its dedicated state demands, that surplus cannot be redirected—it remains idle even if adjacent states face shortages and are drawing on more expensive plants. This is not a market failure in the sense of irrational behavior; it is an equilibrium consequence of the contracting arrangement, where long-term supply security motivates exclusivity terms. The practical implication is that first-best merit-order dispatch is not achieved even absent coal subsidies. Coal market distortions compound this inefficiency: by artificially reducing the private cost of high-emission public plants, they further entrench their position in the dispatch order, amplifying the welfare losses relative to the undistorted counterfactual (Jha et al. (2021)).

5.2.3 Model Validation

The model’s inputs: available capacities $\bar{q}_{i,t}$ and marginal costs $c_{i,t}$, are directly observed or constructed from primary data, as described in Section 5.1. I validate the model against daily plant-level generation for over 200 plants and more than 1,000 days. The model explains 82% of the variation in actual output at the daily plant level ($R^2 = 0.82$), demonstrating that the institutional constraints captured by the optimization—PPA exclusivity, Stage 1 withholding decisions, and national merchant merit-order dispatch—are the primary determinants of observed generation patterns. The model correctly replicates relative utilization rates by ownership type (public versus private) and captures seasonal generation variation. As an additional validation of cost inputs, I compare constructed variable costs against costs that plants

self-report to grid operators under their PPAs (from the MERIT database); for a consistent sample excluding lignite plants and those heavily dependent on imported coal, the correlation between constructed and reported variable costs is 0.81. Since the model contains no free parameters fitted to generation outcomes, this fit reflects the adequacy of the institutional representation, not sample overfitting. The remaining variation reflects hour-to-hour adjustments, strategic outage reporting, and non-cost dispatch that the daily model does not attempt to capture.

Limitations. The model abstracts from technical ramping constraints that prevent instantaneous output adjustment, hour-to-hour variation in transmission capacity (the model operates at daily frequency), and strategic behavior in availability reporting or PPA renegotiation. It also treats transmission as uncongested for non-PPA capacity, which is supported by Central Electricity Authority data showing transmission-related outages constitute less than 1% of total capacity outages during the study period, and by evidence of no meaningful congestion for short-term bilateral and exchange-traded power during our study period ([Athawale and Felder \(2023\)](#)). These abstractions are standard for daily-frequency structural dispatch models ([Fabrizio et al. \(2007\)](#); [Cicala \(2022\)](#)). The primary aim is not high-temporal-resolution dispatch prediction but production reallocation across plants under alternative cost structures—the validation target relevant for counterfactual credibility—for which daily-level fit at $R^2 = 0.82$ is appropriate.

5.3 Counterfactual Analysis

Using this model, I conduct two sets of counterfactual analyses to quantify both the environmental costs of coal market distortions and their impact on the effectiveness of carbon taxes. Throughout these counterfactuals, the model maintains the core institutional features while allowing thermal plants to adjust generation in response to changing marginal costs. The analysis is short-term, abstracting away from dynamic responses such as electricity demand adjustments, generator exit, power plant upgrades, or investment in clean technology like solar or wind. While these dynamic factors are relevant in the medium to long run, the static distortions we identify directly influence investment incentives, as explained in Section 4.4. Therefore, while our quantitative results focus on short-run effects, they have important implications for long-run outcomes.

The first counterfactual examines equilibrium outcomes under current versus undis-

torted coal prices. Under the undistorted scenario, all plants face the CIL/SCCL notified price of coal corresponding to their location and the grade (gross calorific value) of coal they consume. This eliminates the artificial cost advantage currently enjoyed by public sector plants through preferential allocation of subsidized domestic coal. By comparing generation patterns, costs, and emissions, both CO_2 and $PM_{2.5}$ across these scenarios, I quantify the environmental costs of existing market distortions.

To estimate plant-specific emissions, I combine multiple data sources. For CO_2 emissions, I multiply each plant’s heat rate by coal composition. Plant-level gross calorific values come from CEA¹², while carbon content values use production-weighted averages from Srinivasan et al. (2018) for domestic coal and source-specific values for imported coal. For local pollution impacts, I utilize $PM_{2.5}$ mortality intensities from Singh et al. (2024), which account for both plants’ emission rates and atmospheric transport patterns to identify population exposure.

The analysis assumes perfectly inelastic demand set to observed levels. This assumption reflects institutional reality: in the short-run wholesale market we simulate, a fixed set of generators produces electricity to meet daily demand, while end-users face regulated tariffs adjusted only annually. While there is some demand elasticity in power exchanges, this represents less than 1 percent of total energy demand, making demand estimation unnecessary for our analysis.

The second set of counterfactuals evaluates these outcomes under varying carbon tax rates from \$0 to \$100 per tonne of CO_2 . I run these carbon tax simulations under both distorted and undistorted coal markets. This allows me to test the theoretical prediction that input market distortions reduce the effectiveness of carbon taxes by insulating high-emission plants from carbon price signals through preferential fuel allocation. The results, presented in Section 6, strongly support this prediction.

6 Results

In this section, I present results focusing on the post-Covid period from July 2021 to December 2023. This is the period in which coal market distortions became particularly binding. Before this period, two factors effectively nullified the distortions:

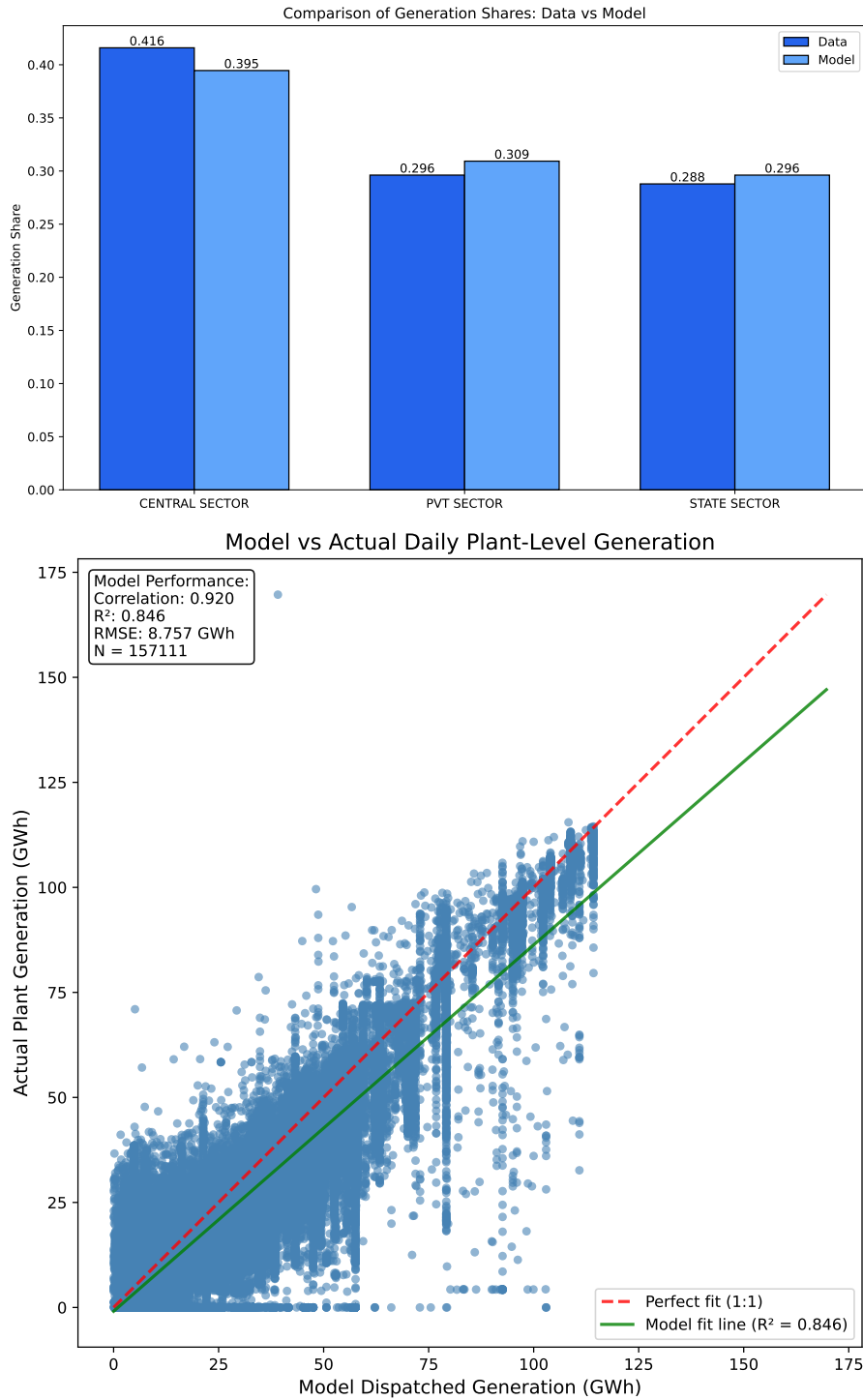
¹²Dataset compiled by Central Electricity Authority (CEA) containing CO_2 emissions data for all grid-connected power stations in India: <https://cea.nic.in/cdm-co2-baseline-database/?lang=en>

historically low electricity demand due to Covid-19 restrictions meant domestic coal was sufficient for most generation needs, while record low international coal prices provided a competitive alternative for plants requiring imports. Post-July 2021, the divergence between private and public plant costs widened substantially as demand recovered and international prices rose sharply, causing private plants' PLF to fall 5 percentage points below public plants despite a 13.5 percent increase in overall electricity demand (Figure A3). The demand patterns observed during this period better reflect both current market conditions and projected near-term trends in India's power sector ([International Energy Agency \(2023\)](#)). This makes the results more relevant for understanding ongoing and future inefficiencies in coal allocation.

6.1 Model fit

The empirical model generates sharp predictions for the daily production decisions of generators. I check the fit along two important moments - aggregate generation shares by ownership, and high resolution moment for daily plant-level absolute generation. Figure 11 compares the model predictions with actual data. The model explains 85 percent of the observed variation in daily production at the power plant level ($R^2 = 0.85$), with a correlation of 0.92 between predicted and actual generation. Model-generated ownership-wise shares stack near-perfectly with the data: as shown in Figure 11, central sector generation shares (39.5% predicted vs. 41.6% actual), state sector (29.6% vs. 28.8%), and private sector (30.9% vs. 29.6%) demonstrate excellent model accuracy with differences under 2 percentage points. For reasons explained previously, the entire generation profile plausibly satisfies the transmission constraints. These statistical metrics are particularly impressive given the complexity of India's power system, which involves multiple regions, varying demand patterns, and diverse generation sources.

Figure 11. Comparison of Generation Shares: Data vs Model



Top panel shows aggregate generation shares by ownership type (central, state, private) comparing model predictions to actual data. Bottom panel shows scatter plot of daily plant-level generation (GWh) comparing model predictions (x-axis) to observed generation (y-axis), with the 45-degree line indicating perfect prediction.

The model demonstrates strong performance across multiple validation criteria. Statistically, it achieves high predictive accuracy for daily plant-level generation decisions. Institutionally, it replicates the observed market structure with 82.4% of modeled electricity flows occurring within PPA obligations, closely matching the actual rate of 88% ([Central Electricity Regulatory Commission \(CERC\)](#)). Operationally, all solutions respect physical transmission constraints. This multi-dimensional validation, covering statistical accuracy, institutional constraints, and operational feasibility, establishes the model’s credibility for counterfactual policy analysis.

6.2 Environmental Costs of Market Distortions

Removing coal market distortions generates substantial environmental benefits through the reallocation of production from inefficient public to efficient private generators. I quantify these benefits in terms of both carbon emissions and local air pollution mortality.

The reallocation of generation across ownership types is stark. As shown in the left panel of Figure 12, private power plants increase generation by 23 percent relative to the status quo. This expansion is offset by corresponding declines in public sector generation, with state-owned plants experiencing the largest reduction (-20 percent), while central government plants see a modest decline (-3.3 percent). These changes reflect the removal of artificial cost advantages that previously kept efficient private plants operating below capacity.

This reallocation drives significant reductions in carbon emissions. The right panel of Figure 12 shows that annual aggregate CO_2 emissions decline by 15 million tons, approximately 1.4 percent of India’s total power sector emissions in this period. While emissions from private generators increase due to their expanded production, this is more than offset by substantial reductions from public plants - 98 million tons from state generators and 23 million tons from central sector plants. To put these reductions in perspective, they equal 4 percent of total U.S. coal power emissions ([U.S. Energy Information Administration \(EIA\)](#)) and 10 percent of India’s entire steel sector emissions ([Garg et al. \(2023\)](#)), the second largest industrial emitter after electricity. At the U.S. EPA’s 2023 social cost of carbon estimate of \$190 per ton, these reductions represent annual welfare gains of \$3.9 billion. Notably, these substantial gains emerge purely from removing the preexisting misallocation in production, without requiring costly investments in clean technology or plant upgrades. Moreover,

the shift toward efficient private generation improves dynamic incentives for future investments in emissions control and efficiency improvements by rewarding cleaner production technology.

The mortality impacts from local air pollution show even larger relative improvements. Using plant-specific mortality intensities from [Singh et al. \(2024\)](#), who employ the InMap atmospheric transport model to estimate $PM_{2.5}$ exposure and mortality, we find that removing coal market distortions would reduce annual deaths by approximately 4,000, or 6 percent. The magnitude of these reductions is large - they equal 25 percent of all deaths associated with fine particulate air pollution from electricity generation in the United States ([Thind et al. \(2019\)](#)). That such large mortality reductions can be achieved through pure market reform, without any explicit environmental regulation, underscores the severe human costs of existing market distortions. Figure [A4](#) reveals substantial heterogeneity in plant-level mortality intensity (deaths per GWh), ranging from 0.002 to 0.43 deaths per GWh. The larger percentage reduction in mortality compared to carbon emissions (6 versus 1.4 percent) reflects the spatial reallocation of production away from high-mortality public plants (shown as large blue circles) toward private generators with more favorable locations relative to population centers. At a conservative value of statistical life of \$1 million, the reduction in mortality alone generates annual welfare gains of \$4 billion. Combined with the carbon benefits, the total annual environmental welfare gain amounts to \$7.9 billion. These estimates suggest that market distortions impose significant environmental costs beyond their direct economic inefficiencies.

Figure 12. Generation and Emissions Changes from Removing Coal Market Distortions

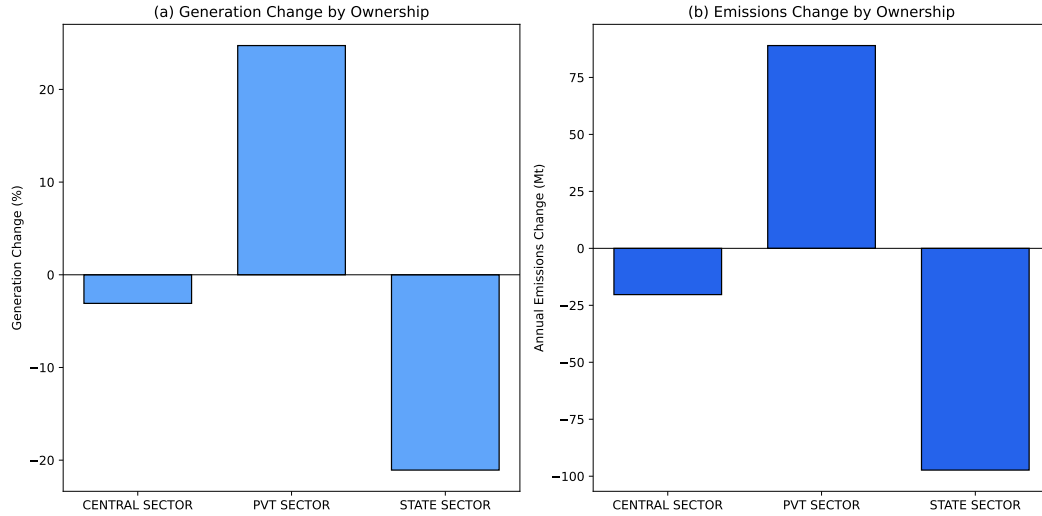


Figure shows the changes in generation (Panel A) and emissions (Panel B) by ownership type after equalizing coal prices across generators. Generation changes are measured as percentage changes from baseline levels, while emissions changes are reported in million tons (Mt) of CO_2 per year. Central Sector refers to plants owned by the central government, State Sector to plants owned by state governments, and PVT Sector to privately-owned plants.

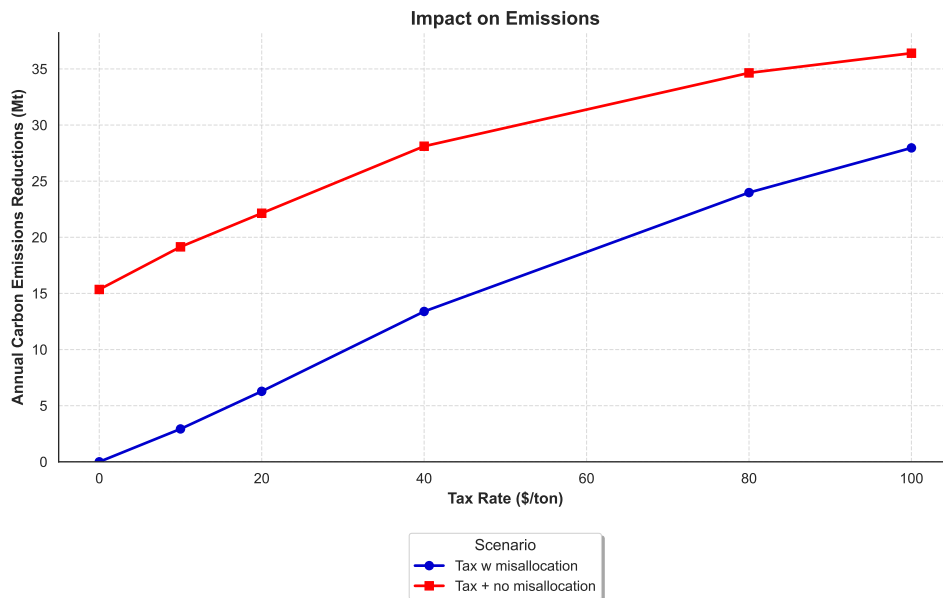
6.3 How distortions impact the effectiveness of carbon tax

This section quantifies how misallocation creates large additional climate damages by systematically undermining environmental policy targeting. Figure 13 compares two scenarios: a carbon tax applied to the current distorted market (blue line), and when reforms that eliminate these distortions accompany carbon taxation (red line).

The first key finding is that coal market reform alone achieves substantial emission reductions without carbon pricing. As shown and discussed in the previous subsection, even at a zero tax rate, removing distortions reduces emissions by 15 million tons CO_2 annually, equivalent to what a \$40/ton carbon tax would achieve in the currently distorted market. This reduction, representing 1.4 percent of India's power sector emissions, emerges purely from allowing more efficient private generators to increase their market share, without any explicit climate policy intervention. The emissions in the highest tax bracket (\$100/ton) correspond to a 3 percent reduction in emissions. The estimated reductions under different carbon tax rates are consistent with estimates reported in literature for Indian electricity sector - [Sengupta et al. \(2022b\)](#) find reductions in emissions intensity of 4 percent for a \$100/ton carbon

tax.

Figure 13. The Impact of Carbon Taxes Under Distorted and Competitive Coal Markets



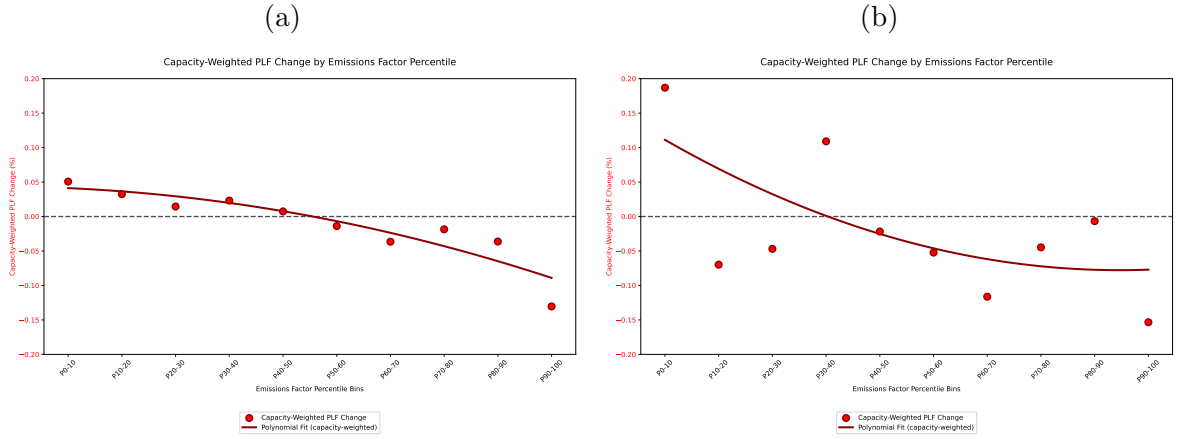
This figure plots annual carbon emissions reductions in million tons (Mt) against carbon tax rates (\$/ton). The blue line shows emissions reductions under current distorted coal markets, while the red line shows reductions when reforms that eliminate these distortions (or equalizing coal prices across generators) accompany carbon taxation. Carbon tax rates vary from \$0-100/ton. Emissions are calculated using plant-level emissions factors, discussed in section 5.3.

The second finding reveals how carbon price signals are muted under preexisting coal market distortions. I find that at every tax rate from \$20-100/ton, if the misallocation from coal distortion was eliminated, carbon taxes would achieve 130-270% greater emissions reductions compared to carbon taxes alone in the status quo. Even at a large tax rate of \$100/ton, carbon taxes yield 28 million tons of reductions compared to 36 million tons if the misallocation from coal market distortions was removed.

These differences in tax impacts operate through specific channels visible in Figure 14. Panel (a) shows that a \$20/ton carbon tax under distorted markets achieves minimal reallocation - the cleanest generators (P0-10 percentile) experience little increase in capacity utilization, while the dirtiest generators (P90-100) see only modest reductions. In stark contrast, panel (b) reveals that removing coal allocation distortions without any carbon tax creates substantial reallocation from dirty to clean plants. The cleanest generators increase their capacity factors by approximately 15 percentage points, while the dirtiest generators experience reductions of 8-10 percent-

age points. This represents a reallocation of approximately 20% of production from the highest-emission plants to the lowest-emission plants under distortion removal, compared to essentially no meaningful reallocation under carbon taxation in the distorted market. This limited reallocation under carbon taxation occurs because fuel cost advantages dominate carbon price signals - public plants' access to subsidized coal at one-third the market price offsets much of the carbon tax's impact on relative costs, preventing the efficient targeting of dirty generators that carbon pricing is designed to achieve.

Figure 14. Production Reallocation under Different Policies



This figure shows the change in capacity-weighted plant load factor (PLF) by emissions factor percentile, where P0-10 represents the cleanest generators and P90-100 represents the dirtiest generators. Panel (a) shows the impact of a \$20/ton carbon tax imposed on the existing distorted system. Panel (b) shows the impact of removing coal allocation distortions without any carbon tax. The red dots indicate the capacity-weighted average change in PLF for each emissions percentile, and the red line shows the polynomial fit across percentiles. Positive values indicate increased capacity utilization, while negative values indicate decreased utilization.

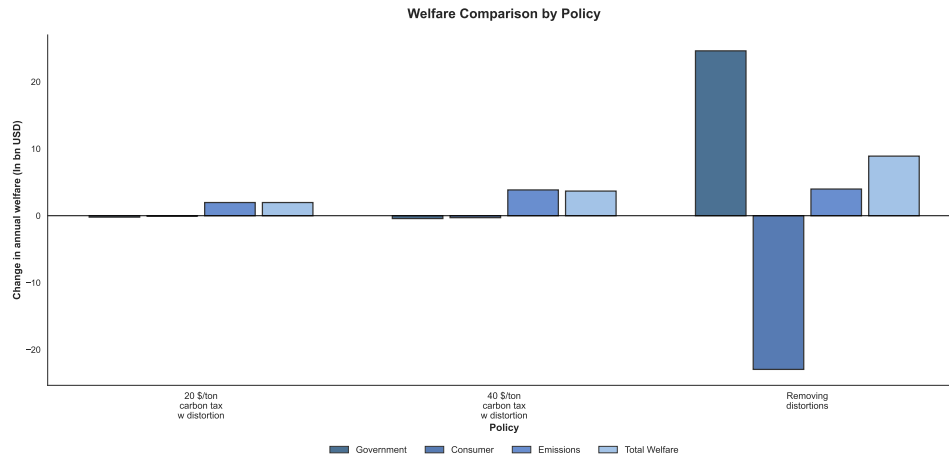
The economic implications of this dampened tax impact are substantial. To achieve the same emission reductions as a \$40/ton tax that accompanies removing the misallocation, tax imposed on current distortions would require a rate of \$100/ton - more than double in magnitude. At a social cost of carbon of \$190/ton, this inefficiency represents additional climate damages of \$2.7 billion annually (14 million tons $\times$ \$190/ton). These excess emissions persist because the carbon tax alone cannot overcome the artificial cost advantages created by preferential coal allocation.

The distortions also create specific barriers to low-cost abatement opportunities. For example, private plants in our sample average heat rates 10-15% better than public plants of similar vintage, implying potential emission reductions of 0.10-0.15 tons CO_2

per MWh of reallocated generation. However, under distorted markets, a \$20/ton carbon tax induces only about 14% of the generation reallocation achieved through market reform, leaving substantial low-cost abatement opportunities untapped.

The welfare analysis reveals that removing coal allocation distortions dramatically outperforms carbon taxation on economic grounds. As shown in Figure 15, coal market deregulation generates total welfare gains of approximately \$8.9 billion annually, compared to \$3.7 billion under a \$40/ton carbon tax with existing distortions—representing welfare gains 2.4 times larger. This superior performance stems from multiple channels: removing distortions generates substantial government revenue gains of \$22 billion through higher Coal India Limited revenues, while carbon taxation under distorted markets produces low government revenue and actually reduces consumer surplus due to higher electricity costs. The environmental benefits are comparable across policies (\$4.6 billion for distortion removal vs \$3.9 billion for the \$40/ton carbon tax), but distortion removal achieves environmental benefits while simultaneously delivering massive efficiency improvements that carbon taxation cannot capture. Even a more aggressive \$40/ton carbon tax generates only modest welfare gains of \$3.7 billion, primarily through environmental benefits, while failing to address the underlying productive inefficiencies. This stark difference demonstrates that when markets are severely distorted, policies that restore efficient allocation can deliver environmental benefits comparable to carbon pricing while generating substantially larger overall welfare gains through improved resource allocation.

Figure 15. Welfare effects under counterfactual environmental policies



The figure shows the change in annual welfare (in billion USD) under three policy scenarios: a \$20/ton carbon tax with existing distortions, a \$40/ton carbon tax with existing distortions, and removing coal allocation distortions without carbon taxation. Welfare changes are decomposed into four components: government revenue effects (dark blue), consumer surplus (medium blue), environmental benefits from emissions reductions valued at the social cost of carbon (blue), and total welfare change (light blue). Government effects capture changes in Coal India Limited revenues. Consumer surplus reflects changes in electricity costs net of the carbon tax revenue transfers. Environmental benefits are calculated using emissions reductions valued at \$190/ton CO_2 . Total welfare represents the sum of all components. Positive values indicate welfare gains, while negative values indicate welfare losses relative to the status quo.

These findings demonstrate that market reform could be an essential complement or even substitute for carbon pricing in developing countries where policy enforcement is challenging. The current focus on carbon taxation, while important, may deliver far less environmental benefit than expected when implemented in distorted markets. Moreover, the significant emissions reductions achievable through market reform alone - equivalent to a moderate carbon tax - suggest that improving market efficiency could be a valuable first step toward addressing climate challenges, particularly in contexts where carbon taxation faces significant opposition.

7 Conclusion

In this paper, I demonstrated that reforms that address pre-existing market distortions can outperform carbon pricing—the consensus climate policy solution—in heavily distorted economies. Using India’s electricity sector, I show that making coal markets competitive achieves CO_2 reductions equivalent to a \$40/tonne carbon tax while generating welfare gains 2.4 times larger. This finding suggests that the se-

quencing of reforms matters fundamentally for environmental outcomes in developing countries.

The results have immediate policy relevance as developing countries design climate policies. First, efficiency reforms should precede carbon pricing: when input subsidies systematically protect dirty producers, carbon taxes may accelerate clean firm exit while leaving dirty incumbents unaffected. Second, the gains from coordinated reform are substantial—eliminating distortions would make carbon taxes 130-270% more effective across a wide range of tax rates. Third, politically sensitive reforms remain feasible: coal price deregulation coupled with direct consumer compensation can achieve Pareto improvements, demonstrating that efficiency and distributional concerns need not conflict.

These findings extend beyond India’s electricity generation industry to any setting where administrative allocation or state enterprise favoritism alters the relationship between productivity and social costs—common features across developing countries in capital-intensive, high-polluting industries. As countries like India implement Carbon Credit Trading Schemes and other market-based environmental policies, understanding how pre-existing distortions affect policy design becomes crucial for global climate efforts.

More broadly, my findings bridge the misallocation and environmental economics literature in developing country settings by showing that production distortions have environmental consequences beyond their well-documented productivity costs. The finding that efficiency reforms can serve as environmental policy instruments suggests a fundamental rethinking of climate policy design in developing economies: the path to decarbonization may require fixing markets first, then pricing carbon.

References

- Acemoglu, Daron, Vasco M. Carvalho, Asuman Ozdaglar, and Alireza Tahbaz-Salehi**, “The Network Origins of Aggregate Fluctuations,” *Econometrica*, 2012, 80 (5), 1977–2016. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA9623>.
- Athawale, Rasika and Frank A Felder**, “Overbuilding transmission: A case study and policy analysis of the Indian power sector,” *Energy Policy*, 2023, 174, 113437.
- Barrot, Jean-Noël and Julien Sauvagnat**, “Input Specificity and the Propagation of Idiosyncratic Shocks in Production Networks *,” *The Quarterly Journal of Economics*, August 2016, 131 (3), 1543–1592.
- Baylis, Patrick, Michael Greenstone, Kenneth Lee, and Harshil Sahai**, “Is the Demand for Clean Air Too Low? Experimental Evidence from Delhi,” November 2023. Working Paper.
- Bergquist, Lauren F, Danial Lashkari, and Eric Verhoogen**, “Wedges: A Microeconomic Perspective on Misallocation,” Technical Report, National Bureau of Economic Research 2026.
- Berkouwer, Susanna B. and Joshua T. Dean**, “Credit, Attention, and Externalities in the Adoption of Energy Efficient Technologies by Low-Income Households,” *American Economic Review*, October 2022, 112 (10), 3291–3330.
- Bovenberg, A Lans and Lawrence H Goulder**, “Optimal environmental taxation in the presence of other taxes: general-equilibrium analyses,” *The American Economic Review*, 1996, 86 (4), 985–1000.
- Bureau of Energy Efficiency**, “Detailed Procedure for Compliance Mechanism under CCTS,” 2023.
- Central Electricity Authority**, “Daily Generation Report,” Technical Report, Central Electricity Authority 2018-2023. Daily Reports.
- , “Monthly Coal Report,” Technical Report, Central Electricity Authority 2018-2023. Monthly Reports.
- Central Electricity Regulatory Commission (CERC)**, “Annual Report 2021-22,” Technical Report 2022.
- Cicala, Steve**, “Imperfect Markets versus Imperfect Regulation in US Electricity Generation,” *American Economic Review*, February 2022, 112 (2), 409–441.
- Comptroller and Auditor General of India**, “Performance Audit Report on Allocation of Coal Blocks and Augmentation of Coal Production,” Performance Audit Report 7 2012.
- Cropper, Maureen, Ryna Cui, Sarath Guttikunda, Nate Hultman, Puja Jawahar, Yongjoon Park, Xinlu Yao, and Xiao-Peng Song**, “The mortality

- impacts of current and planned coal-fired power plants in India,” *Proceedings of the National Academy of Sciences*, 2021, *118* (5), e2017936118.
- Davis, Lucas W. and Catherine Wolfram**, “Deregulation, Consolidation, and Efficiency: Evidence from US Nuclear Power,” *American Economic Journal: Applied Economics*, July 2012, *4* (4), 194–225.
- Dollar, David and Shang-Jin Wei**, “Das (Wasted) Kapital: Firm Ownership and Investment Efficiency in China,” May 2007.
- Domínguez-Iino, Tomás**, “Efficiency and Redistribution in Environmental Policy: An Equilibrium Analysis of Agricultural Supply Chains,” 2024. Working Paper.
- Duflo, Esther, Michael Greenstone, Rohini Pande, and Nicholas Ryan**, “Truth-telling by third-party auditors and the response of polluting firms: Experimental evidence from India,” *The Quarterly Journal of Economics*, 2013, *128* (4), 1499–1545.
- , –, –, and –, “The Value of Regulatory Discretion: Estimates From Environmental Inspections in India,” *Econometrica*, 2018, *86* (6), 2123–2160. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA12876>.
- Fabrizio, Kira R., Nancy L. Rose, and Catherine D. Wolfram**, “Do Markets Reduce Costs? Assessing the Impact of Regulatory Restructuring on US Electric Generation Efficiency,” *American Economic Review*, September 2007, *97* (4), 1250–1277.
- Garg, Vibhuti, Jyoti Gulia, Kapil Gupta, Nagoor Shaik, and Shantanu Srivastava**, “Steel Decarbonisation in India,” Technical Report, Institute for Energy Economics and Financial Analysis 2023.
- Greenstone, Michael and B. Kelsey Jack**, “Envirodevonomics: A Research Agenda for an Emerging Field,” *Journal of Economic Literature*, March 2015, *53* (1), 5–42.
- and **Rema Hanna**, “Environmental Regulations, Air and Water Pollution, and Infant Mortality in India,” *American Economic Review*, October 2014, *104* (10), 3038–3072.
- Hsieh, Chang-Tai and Peter J. Klenow**, “Misallocation and Manufacturing TFP in China and India*,” *The Quarterly Journal of Economics*, November 2009, *124* (4), 1403–1448.
- International Energy Agency**, “World Energy Outlook 2023,” Technical Report, IEA, Paris 2023. Licence: CC BY 4.0 (report).
- Jain, Purva**, “Meeting peak power demand through gas-based plants requires dynamic domestic resource allocation,” Technical Report, Institute for Energy Economics and Financial Analysis 2024.

- Jayachandran, Seema**, “How Economic Development Influences the Environment,” *Annual Review of Economics*, 2022, 14 (1), 229–252. eprint: <https://doi.org/10.1146/annurev-economics-082321-123803>.
- Jha, Akshaya, Louis Preonas, and Fiona Burlig**, “Blackouts: The role of India’s Wholesale Electricity Market,” Working Paper 29610, National Bureau of Economic Research December 2021.
- Kumar, Alok and Sushanta Chatterjee**, *Electricity Sector in India: Policy and Regulation*, Oxford University Press, October 2012.
- Lipsey, R. G. and Kelvin Lancaster**, “The General Theory of Second Best,” *The Review of Economic Studies*, 1956, 24 (1), 11–32. Publisher: [Oxford University Press, Review of Economic Studies, Ltd.].
- Mathavan, Deeptha**, “From Dabhol to Ratnagiri: The electricity act of 2003 and reform of India’s power sector,” *Colum. J. Transnat’l L.*, 2008, 47, 387.
- Ministry of Coal**, October 2007. New Coal Distribution Policy.
- , “Annual Report,” Technical Report, Ministry of Coal 2014.
- , “SHAKTI (Scheme for Harnessing and Allocating Koyala Transparently in India) Policy,” Policy Notification 23011/15/2016-CPD/CLD, Government of India May 2017.
- Ministry of Power**, “National Tariff Policy, Resolution No. 23/2/2005 (III),” *Gazette of India*, 2006, pp. 389, 415.
- Nordhaus, William D.**, “Reflections on the Economics of Climate Change,” *Journal of Economic Perspectives*, December 1993, 7 (4), 11–25.
- Oliva, Paulina**, “Environmental regulations and corruption: Automobile emissions in Mexico City,” *Journal of Political Economy*, 2015, 123 (3), 686–724.
- Pigou, Arthur Cecil**, *The Economics of Welfare (4th ed.)*, London: Macmillan, 1932, 1932.
- Preonas, Louis**, “Market Power in Coal Shipping and Implications for U.S. Climate Policy,” *The Review of Economic Studies*, July 2024, 91 (4), 2508–2537.
- Restuccia, Diego and Richard Rogerson**, “The Causes and Costs of Misallocation,” *Journal of Economic Perspectives*, August 2017, 31 (3), 151–174.
- Ritchie, Hannah, Pablo Rosado, and Max Roser**, “Greenhouse gas emissions,” *Our World in Data*, 2020. <https://ourworldindata.org/greenhouse-gas-emissions>.
- Ryan, Nicholas**, “Contract enforcement and productive efficiency: Evidence from the bidding and renegotiation of power contracts in India,” *Econometrica*, 2020, 88 (2), 383–424.

- , “Contract Enforcement and Productive Efficiency: Evidence From the Bidding and Renegotiation of Power Contracts in India,” *Econometrica*, 2020, 88 (2), 383–424. eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA17041>.
- , “The Competitive Effects of Transmission Infrastructure in the Indian Electricity Market,” *American Economic Journal: Microeconomics*, May 2021, 13 (2), 202–42.
- , “The Competitive Effects of Transmission Infrastructure in the Indian Electricity Market,” *American Economic Journal: Microeconomics*, May 2021, 13 (2), 202–242.
- Safiullah, Hameed, Gabriela Hug, and Rahul Tongia**, “Design of load balancing mechanism for Indian electricity markets,” *Energy Systems*, 2017, 8, 309–350.
- Sehgal, Anurag and Rahul Tongia**, “Coal Pricing and Allocations in India,” in Rahul Tongia, Anurag Sehgal, and Puneet Kamboj, eds., *Future of Coal in India*, Brookings India, 2020.
- Sengupta, Shayak, Peter J. Adams, Thomas A. Deetjen, Puneet Kamboj, Swati D’Souza, Rahul Tongia, and Inês M. L. Azevedo**, “Subnational implications from climate and air pollution policies in India’s electricity sector,” *Science*, 2022, 378 (6620), eabh1484.
- , **Sumil K. Thakrar, Kirat Singh, Rahul Tongia, Jason D. Hill, Ines M. L. Azevedo, and Peter J. Adams**, “Inequality in air pollution mortality from power generation in India,” *Environmental Research Letters*, December 2022, 18 (1), 014005. Publisher: IOP Publishing.
- Shapiro, Joseph S**, “The environmental bias of trade policy,” *The Quarterly Journal of Economics*, 2021, 136 (2), 831–886.
- Singh, Kirat, Tapas Peshin, Shayak Sengupta, Sumil K Thakrar, Christopher W Tessum, Jason D Hill, Inês M L Azevedo, and Stephen P Luby**, “Air pollution mortality from India’s coal power plants: unit-level estimates for targeted policy,” *Environmental Research Letters*, May 2024, 19 (6), 064016. Publisher: IOP Publishing.
- Srinivasan, S., N. Roshna, S. Guttikunda, A. Kanudia, S. Saif, and J. Asundi**, “Benefit Cost Analysis of Emissions Standards for Coal-Based Thermal Power Plants in India,” Technical Report CSTEP report 2018-06, Center for Study of Science, Technology and Policy 2018.
- Taneja, Mansi**, “Indonesia’s coal pricing norms hit Tata’s Mundra UMPP,” *Business Standard*, Jan 2013.
- Thind, Maninder P. S., Christopher W. Tessum, Inês L. Azevedo, and Julian D. Marshall**, “Fine Particulate Air Pollution from Electricity Generation in the US: Health Impacts by Race, Income, and Geography,” *Environmental Sci-*

ence & Technology, December 2019, 53 (23), 14010–14019. Publisher: American Chemical Society.

U.S. Energy Information Administration (EIA), “State Electricity Profiles,” *U.S. Energy Information Administration (EIA)*.

Xia, Fan and Bing Zhang, “Pollution Taxes as a Second-Best: Accounting for Multidimensional Firm Heterogeneity in Environmental Regulations,” 2023. Working Paper.

A Appendix Tables and Figures

Appendix Figure A1. Electricity generation by source in major countries

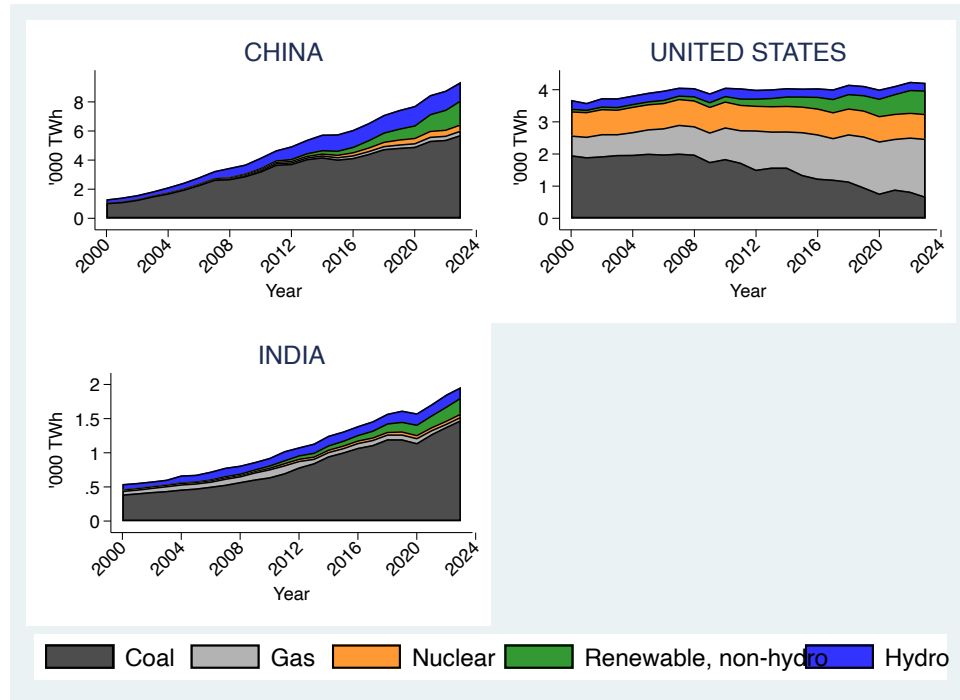


Figure shows electricity generation by source in China, United States, and India from 2000-2024. Generation is measured in thousand terawatt-hours (TWh) and broken down into five categories: coal (dark gray), gas (light gray), nuclear (orange), non-hydro renewables (green), and hydro (blue). Source: Ember (2024); Energy Institute - Statistical Review of World Energy (2024)

Appendix Figure A2. Plant Heat Rate by Ownership and Commissioning Year

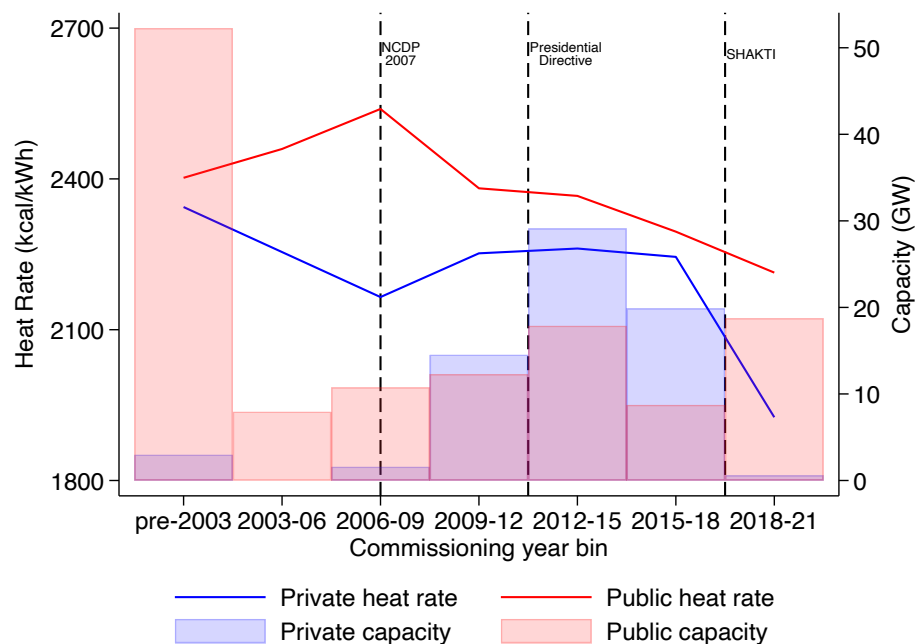


Figure shows plant heat rates (in kcal/kWh) by ownership and vintage. The graph plots heat rate against plant commissioning year for public (red) and private (blue) generators. Lines show average heat rate by ownership while bars show installed capacity (GW, right axis) in each vintage bin. Vertical dashed lines mark key policy changes: New Coal Distribution Policy (2007), Presidential Directive on fuel supply agreements (2012), and SHAKTI policy (2017), explained in detail in section 2.

Appendix Figure A3. Private-Public Differences in Marginal Costs and PLF Over Time

he figure plots average monthly differences in marginal costs and plant load factor

Appendix Figure A4. Power Plant Mortality Intensities

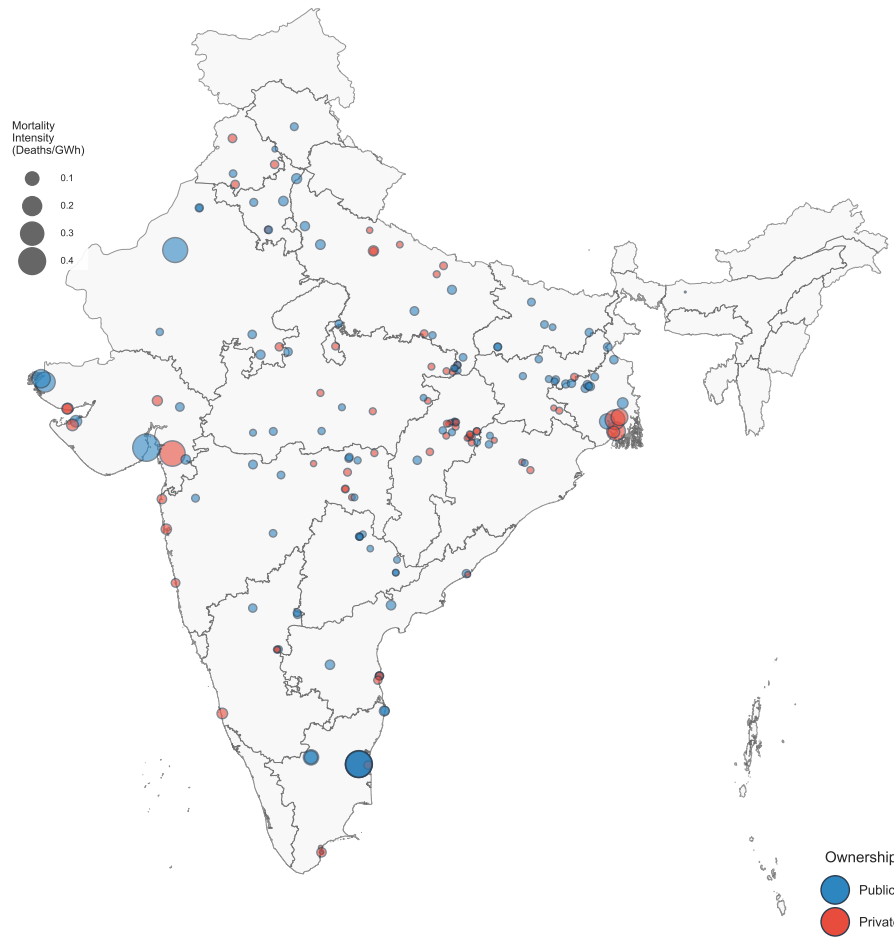


Figure shows the geographical distribution of coal based power plants in India and their mortality intensity, expressed in Deaths/GWh of generation. The red and blue circles represent privately and publicly owned power plants respectively. The size of the circles are in proportion of the plants' mortality intensity. Source: [Singh et al. \(2024\)](#)

B Appendix One: Variable cost estimation

The methodology for estimating plant-specific variable cost of electricity generation (VC), expressed in INR per kWh, is based on a formula provided by the Central Electricity Regulatory Commission (CERC) Tariff Regulations 2019. The formula is as follows:

$$VC = \left(\frac{(SHR - SFC \times CVSF) \times LPPF}{CVPF} + SFC \times LPSFi \right) / (1 - AUX)$$

where:

SHR = Gross station heat rate, in kCal per kWh

SFC = Normative specific fuel oil consumption, in ml per kWh

$CVSF$ = Calorific value of secondary fuel, in kCal per ml

$LPPF$ = landed fuel cost of primary fuel, in INR per kg or cubic metre, as applicable

$CVPF$ = Gross calorific value of primary fuel, in kCal per kg or cubic metre, as applicable

$LPSFi$ = Landed cost of secondary fuel in INR per ml

AUX = Normative auxiliary energy consumption

The formula incorporates various factors that contribute to the variable cost of electricity generation, including the efficiency of the power plant (represented by the station heat rate), the costs and calorific values of both primary and secondary fuels, and the energy consumed by the plant's auxiliary systems. The CERC Tariff Regulations also specify normative assumptions to be used in determining the variable cost. These assumptions vary based on unit capacity, vintage, and type of primary fuel used in the power plant. I use these normative assumptions in the above formula, except for heat rate.^{13 14} I estimate heat rate from data on daily plant-level electricity generation and coal consumption and annually reported average gross calorific value of primary fuel.

¹³The regulations can be found here: <https://www.cercind.gov.in/2019/regulation/Tariff%20Regulations-2019.pdf>

¹⁴Assumed $LPSFi$ = Rs. 50000/kL. This assumption is based on the range of values approved by State Electricity Regulatory Commissions (SERCs) for the period under consideration. These commissions approve the variable costs of power plants annually upon submission of tariff petitions, with approved $LPSFi$ values typically ranging from 45,000 to 60,000 INR per kiloliter. For instance, see https://cserc.gov.in/upload/order/0-9-2023_6878-1692785214-.pdf and <https://www.cercind.gov.in/2024/orders/445-GT-2020.pdf>

The contribution of secondary fuel to the overall variable cost is marginal. The majority of the variable cost is derived from the consumption of the primary fuel, which in this case is coal. I explain the methodology for constructing the landed cost of coal, a crucial component of this calculation, in a separate section.

Comparison with MERIT database

Calculated variable costs of power for coal units closely align with reported variable costs (fig. B1) from the Merit Order Despatch of Electricity for Rejuvenation of Income and Transparency (MERIT), Ministry of Power database ???. This database reports variable generation costs based on long-term power purchase agreements (PPAs) between generators and distribution companies (Discoms). For lignite-based power plants, due to the lack of reliable lignite price data, I assume the reported variable cost from the MERIT India database.

Panel A compares the calculated variable cost of all power plants in the MERIT database for the year 2023 with their reported variable cost, yielding a moderately high correlation of 0.66.

In Panel B, I refine the analysis by eliminating three categories of plants: lignite-based plants, those receiving more than 90 percent of their coal supply from imports, and plants with MERIT variable costs lower than calculated variable costs. This refinement removes 37 out of 134 power plants from Panel A.

It is important to note that the MERIT database doesn't report the "true" variable cost faced by power plants that procure imported coal or buy it at e-auctions. Short-term changes in such fuel costs are not allowed pass-through in the power purchase agreements (PPAs). Consequently, during periods of high market prices for coal, power plants may become economically unviable to operate despite showing "low" variable costs in their PPAs. This discrepancy led to a complex situation. Until the April 11, 2017 Supreme Court judgment, these power plants would claim compensatory tariffs in addition to the PPA tariff to offset high fuel costs. This decision halted this practice, and tariff revisions are now subject to contract renegotiation to amend the PPAs.¹⁵ Given these complexities and the potential mismatch between reported and actual costs, I disregard the MERIT variable cost for these power plants

¹⁵For instance, Mundra TPS signed a supplemental PPA with Haryana after months of renegotiation with a significantly higher tariff <https://www.hindustantimes.com/cities/chandigarh-news/adani-power-agrees-to-sign-supplementary-ppa-with-haryana-101677691257073.html>

in my analysis.

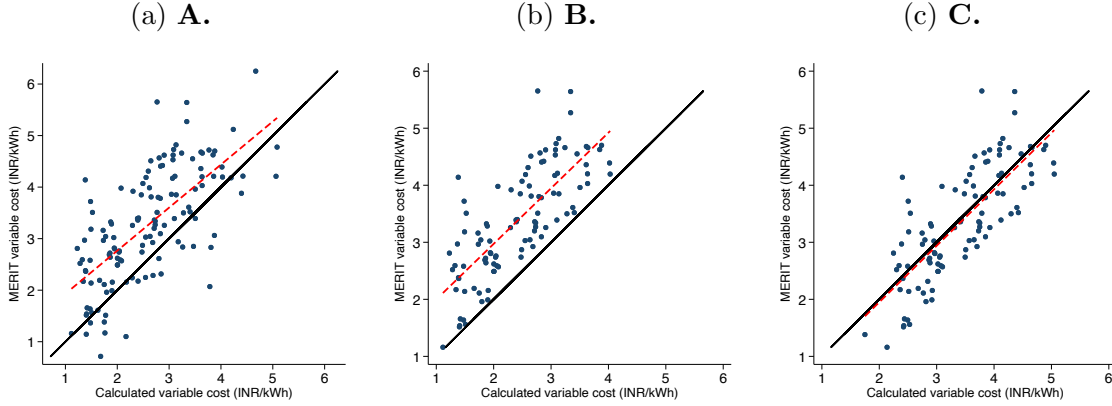
Among the remaining plants, only 7 power plants have calculated variable costs higher than MERIT variable costs, after excluding lignite-based plants and those receiving less than 95 percent of their coal supply domestically. This suggests that, according to MERIT figures, these 7 plants face lower fuel costs than linkage-based power plants. Such a scenario can occur, for example, due to changes in allocated mines towards the end of the study period, as part of ongoing coal cost rationalization efforts.¹⁶ For instance, the data shows that in 2022, towards the end of the study period, the Talwandi Sabo thermal power station in Punjab began sourcing part of its supply from Madhya Pradesh instead of Odisha, significantly reducing its transport costs.

Such re-allocations occurred late in the study period. Therefore, for these 7 power plants, I disregard the variable costs reported in MERIT and continue using the calculated variable costs in subsequent analyses.

After these adjustments, focusing on the remaining 97 power plants yields a very high correlation of 0.81, lending credibility to my calculated variable costs. Finally, as shown in Panel C, I adjust the calculated variable cost of power by 1.03 INR/kWh to close the gap between the lines of best fit and the 45-degree line in Panel B. This adjustment achieves parity between MERIT variable costs and calculated variable costs while preserving the merit order generated by the calculated variable costs. These adjusted variable costs are used throughout the paper for further analysis.

¹⁶See Ministry of Coal's press release for a great summary of these efforts: <https://pib.gov.in/PressReleasePage.aspx?PRID=1953997#:~:text=Four%20Rounds%20of%20Rationalization%20of,642>

Appendix Figure B1. Comparison with MERIT database



Comparison of calculated variable cost of electricity to reported variable cost from Government of India MERIT database with lines of best fit (red dashed) and 45-degree line (black solid). Each point on the graph represents annual average monthly variable cost of electricity generation at one power plant. Panel (A) provides a comparison for all power plants in the MERIT database. Then, panel (B) eliminates lignite based plants, plants receiving greater 90 percent of their coal supply from imports and plants with MERIT variable costs lower than calculated variable costs. Panel (C) further adjusts the comparison by adding 1.03 INR/kWh to the calculated variable costs, closing the gap between line of best fit and 45-degree line in Panel (B).

B.1 Coal Price Construction

B.1.1 Basic Price of Coal

Price of linkage based coal: Power plants with linkage or fuel supply agreements (FSAs) with Coal India Limited (CIL) or Singareni Collieries Company Limited (SCCL) receive coal at administered prices. These prices are notified aperiodically by CIL and SCCL and are reported in rupees per tonne ¹⁷. The methodology for determining the price of linkage based coal involves collecting grade-specific notified prices from both CIL and SCCL. Here, "grade" refers to the calorific value of coal, expressed in kilocalories (kcal) per kg. Each power plant is then assigned a "minemouth" (or "pithead") price based on the average grade of coal it consumes.

E-auction price: Some power plants source a portion of their coal supply from e-auctions conducted monthly by both CIL and SCCL. These e-auctions are conducted under different schemes, with power producers mainly bidding for coal under the spot e-auction and special forward e-auction schemes.

The Spot E-Auction scheme, operational since November 2007, allows any Indian

¹⁷The notified prices of coal are available on the websites of CIL and SCCL.

buyer to procure coal through a consumer-friendly single window for their own consumption or for trading. The Special Forward E-Auction scheme was introduced in 2015-16 to make coal available to power producers with flexible lifting periods.

To construct the e-auction price, I collect data on the average monthly price (reported as a percentage premium over notified price of coal) and the total quantity of coal auctioned by scheme.¹⁸ The average e-auction price for power producers is then calculated as the weighted average of the prices from the two schemes, with weights based on the quantities procured under each. For months with missing data, I use the average scheme-specific price and quantity from the Ministry of Coal's Annual Report.¹⁹ I assume this calculated price to be the "pithead" price for the share of coal procured from e-auctions.

Price of imported coal: There is no plant-specific price data for imported coal in India. To address this, I develop a methodology that leverages available data sources and makes informed assumptions about coal quality, measured in terms of Gross Calorific Value (GCV) and origins. My primary data source is the Directorate General of Commercial Intelligence and Statistics (DGCIS), Ministry of Commerce and Industry which reports monthly quantity and value of imported goods, categorized by country of origin and 8-digit product code. Power plants in this study use steam coal, corresponding to product code 27011920. Notably, approximately 90 percent of this coal is imported from two main sources: Indonesia and South Africa. The quality of coal from these sources differs significantly. I assume that Indonesian coal has an average GCV of 4600 kcal/kg, while South African coal has a higher average GCV of 6000 kcal/kg²⁰.

Using the DGCIS data, I first calculate the monthly average price of imported coal. I then estimate the average GCV of imported coal by taking a weighted average of the GCVs of Indonesian and South African coal. This calculation yields an average GCV of 4900 kcal/kg for imported coal. To validate this estimate, I compare it with the average GCV reported by power plants using imported coal, finding a close match that supports my approach.

¹⁸Data sourced from monthly reports to the Cabinet, available at: <https://coal.gov.in/en/public-information/monthly-report-cabinet>

¹⁹See the chapter "Coal Distribution and Marketing" in the Ministry of Coal's Annual Reports, available at: <https://coal.gov.in/en/public-information/reports/annual-reports>

²⁰See the "Revised Standard Operating Procedure in respect of National Coal Index" published by the Ministry of Coal for the source behind these assumptions, available at: <https://coal.nic.in/sites/default/files/2021-03/05-03-2021a-nci.pdf>

Individual power plants vary in the GCV of the imported coal they use and consequently face different prices. To account for this variation, I next construct a plant-specific GCV price for imported coal. My approach involves expressing the plant-specific reported GCV as a convex combination of the GCVs of Indonesian and South African coal. The weights in this combination are based on the respective country shares in the total imported quantity. For example, if a plant reports using coal with a GCV of 5000 kcal/kg, this implies approximately 71.4 percent of the coal (by weight) came from Indonesia and 28.6 percent from South Africa. The price corresponding to that specific grade is then estimated as the weighted sum of the prices of Indonesian and South African coal, using these same weights.

To ensure the reliability of my estimates, I validate them using a subsample of grade-specific coal import prices sourced from CoalMint. I observe close alignment between my estimates and this independent data source, lending additional confidence in the methodology.

B.1.2 Taxes and surcharge

The basic coal price, as constructed in the previous section, is subject to a range of surcharges, government taxes, and levies. These additional costs vary depending on the source of the coal (linkage-based, e-auction, or imported) and the specific mining company involved.

For linkage-based coal, various surcharges are imposed by Coal India Limited (CIL) and Singareni Collieries Company Limited (SCCL) over the "pithead" price. These surcharges differ between the two companies. For CIL, these include Rs. 270 per tonne on pithead price of Run of Mine (ROM) coal for the supply of steam coal, sizing charges of Rs. 56 per tonne, loading charges of Rs 29 per tonne, and surface transportation charge for transportation of coal from pithead to loading points (assuming 0-3 KM distance) of Rs 40 per tonne. For SCCL, these include Rs. 220 per tonne for the supply of steam coal, sizing charges of Rs. 80 per tonne, loading charges of Rs 60 per tonne, and surface transportation charge of Rs 45 per tonne. SCCL imposes additional charges that vary over time, such as land adjustment fee, forest permit fee, fuel surcharge, explosive cost adjustment, and a charge towards Pension fund.²¹

²¹As of January 1, 2023, SCCL charges include a land adjustment fee of Rs. 61 per tonne, forest permit fee of Rs. 10 per tonne, fuel surcharge of Rs. 686 per tonne, explosive cost adjustment of Rs. 146.4 per tonne, and "Corpus of CMPS 1998 (Pension fund)" charge of Rs. 10 per tonne. See

I add these charges to the notified price and generate what constitutes as the basic price of coal. Additional government taxes and duties are then applied to this basic price.²²

In my calculations, I include a royalty of 14 percent, with the exception of coal sourced from West Bengal, which charges a different royalty for coal sourced from its mines²³. Furthermore, a District Mineral Foundation (DMF) Trust charge of 30 percent on the Royalty and a National Mineral Exploration Trust (NMET) charge of 2 percent on the Royalty are applied. The Goods and Services Tax (GST) is levied at 5 percent on the taxable value, and a GST Compensation Cess of Rs. 400 per tonne of dispatch is also imposed²⁴.

For coal procured through e-auctions, I apply the same set of surcharges, fees, and taxes on the "pithead" price realized in these auctions. A small subset of power plants source all or part of their coal from captive mines. For this supply, I assign the linkage based Coal India Limited price.

Imported coal is subject to a different tax structure. Unlike domestically sourced coal, imported coal is not subject to CIL/SCCL charges or Goods and Services Tax (GST). Instead, the government imposes a GST Compensation Cess of Rs. 400 per tonne and a 2.5 percent import duty on the imported price of coal, which I include in my calculations.

B.1.3 Distance calculation

Finally, to arrive at the plant-specific "delivered price" of coal, transportation charges must be added to the previously estimated coal price. The majority of power plants receive their coal via train, and these transportation charges, assessed in rupees per tonne, vary both over time and by the distance between the mine and the plant. I calculate these transportation charges as follows. First, railway freight rates are col-

details at: https://scclmines.com/scclnew/our-business_coal-prices.asp

²²The additional charges imposed by Coal India Limited over and above the notified price are available at: <https://bcclweb.in/S&M%20Documents/pricing767.pdf>. The surface transportation charges are here [https://www.secl-cil.in/writereaddata/stc%20notice\(64dt18082022\).pdf](https://www.secl-cil.in/writereaddata/stc%20notice(64dt18082022).pdf)

²³For the range of GCV values of coal supplied from West Bengal, the relevant royalty is Rs. 2.5 per tonne. See details at: https://coal.gov.in/sites/default/files/2019-11/180512_0_0.pdf

²⁴The details of taxes, royalty, Goods and Service Tax (GST), Cess, and other levies imposed on coal are available at: <https://sansad.in/getFile/annex/253/AU2088.pdf?source=pqars>

lected from the Indian Railways website. These rates provide the basis for calculating the transportation costs.

To calculate distance by rail between coal plants and coalfields, I combine plant geographic coordinates collected from The World Resources Institute?? with geospatial data on India’s coalfields from the U.S. Geological Survey. These two datasets are then integrated with rail network data created by ML InfoMap to calculate the rail distance between each power plant and its coal source.²⁵

The transportation cost per tonne of coal for each power plant is then calculated by multiplying the estimated distance with the relevant freight rate. This method provides a plant-specific transportation cost that accounts for both the distance traveled and the time-varying railway freight rates. A special case is made for ”pithead” power plants, which are categorized as plants co-located next to a mine with a distance of less than 100 km from the mine. For these plants, the rates of the Merry-Go-Round (MGR) system of Indian Railways are applied. This system, designed for short-distance coal transport, typically offers more favorable rates for these nearby plants. Power plants designated for imported coal are assumed to incur zero surface transportation costs as these are typically located close to seaports.

²⁵More information on these datasets can be found here: <https://earthworks.stanford.edu/catalog/stanford-ww857qy4996>

C Appendix Two